Engineering Mathematics

Detailed Solutions

GATE ENGINEERING MATHEMATICS

PREPFUSION

Previous Year Questions Solutions Series

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Complex Variable Analysis

Engineering Mathematics

Q5.25.1 MSQ 2025 2M Ans: A,B ● ● ○

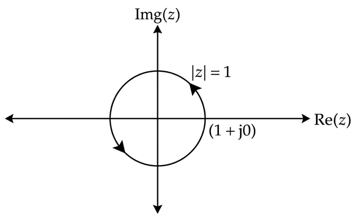

Fig. Solution Q5.25.1

(i) $f(z) = e^z$ is an always analytic function.

$$\therefore \oint_c e^z\, dz = 0$$

(ii) $f(z) = z^n, x \in$ even integer, $f(z)$ is an always analytic function.

$$\therefore \oint_c z^n\, dz = 0$$

(iii) $f(z) = \cos z$ is also an analytic function in contour c.

$$\oint_c f(z)\, dz = 0$$

(iv) $f(z) = \sec z = \frac{1}{\cos z}$

$$\text{Poles: } \cos z = 0$$

$$\Rightarrow z = \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$$

Hence, no poles of $f(z)$ lie inside contour c ($|z| = 1$). Thus, $f(z)$ is analytic inside and on contour c.

$$\therefore \oint_c \sec z\, dz = 0$$

Hence, option (A) and (B) are correct.

$$\boxed{\text{A. } \oint_C e^z\, dz = 0 \text{ , B. } \oint_C z^n\, dz = 0, \text{ where } n \text{ is an even integer}}$$

Key Points

- Exponential, Sinusoidal, and Polynomials in z are always analytic over the entire z-plane (entire functions).
- **Cauchy-Goursat Theorem:** If a function $f(z)$ is analytic at all points interior to and on a simple closed contour c, then $\oint_c f(z)\, dz = 0$.
- Singularities of $\sec z$ occur where $\cos z = 0$, giving $z = \pm\frac{(2n+1)\pi}{2}$. The closest singularities to the origin are at $z = \pm\frac{\pi}{2} \approx \pm 1.57$, which lie entirely outside the unit circle $|z| = 1$.

Q5.24.1 MCQ 2024 2M Ans: D ● ● ○

Given function:

$$f(z) = \frac{\sin(\pi z)}{z^2(z-2)}$$

We need to evaluate $\int_C f(z)\,dz$ along the contour $C : |z| = 3$. The poles of $f(z)$ are given by equating the denominator to zero:

$$z^2(z-2) = 0$$

This gives:

- $z = 0$ is a pole of order 2 (double pole).
- $z = 2$ is a pole of order 1 (simple pole).

Both poles $z = 0$ and $z = 2$ lie inside the contour $C : |z| = 3$. Residue at $z = 0$ (double pole):

$$R_1 = \lim_{z\to 0} \frac{d}{dz}\left[z^2 f(z)\right] = \lim_{z\to 0} \frac{d}{dz}\left(\frac{\sin \pi z}{z-2}\right)$$

Applying the quotient rule for differentiation:

$$= \lim_{z\to 0} \frac{(z-2)\pi\cos\pi z - \sin\pi z}{(z-2)^2}$$

$$= \frac{(0-2)\pi\cos(0) - \sin(0)}{(0-2)^2} = \frac{-2\pi - 0}{4} = -\frac{\pi}{2}$$

Residue at $z = 2$ (single pole):

$$R_2 = \lim_{z\to 2}(z-2)f(z) = \lim_{z\to 2}\frac{\sin(\pi z)}{z^2}$$

$$= \frac{\sin(2\pi)}{2^2} = \frac{0}{4} = 0$$

By Cauchy's Residue Theorem:

$$\therefore \int_C f(z)\,dz = 2\pi i(R_1 + R_2)$$

$$= 2\pi i\left(-\frac{\pi}{2} + 0\right) = -\pi^2 i$$

$$\boxed{\text{D. } -\pi^2 j}$$

Key Points

- **Cauchy's Residue Theorem: If C is a simple closed, positively oriented contour, traversed anticlockwise in complex plane, and $f(z)$ is analytic inside and on C except at finite number of isolated singularities $z_1, z_2, \ldots, z_n$ inside C:**

$$\oint_C f(z)\,dz = 2\pi i \sum_{k=1}^{n} \text{Res}(f, z_k)$$

- **Residue at a Pole of Order m:** The formula for finding the residue at an m-th order pole $z = a$ is:

$$\text{Res}(f, a) = \frac{1}{(m-1)!}\lim_{z\to a}\frac{d^{m-1}}{dz^{m-1}}\left[(z-a)^m f(z)\right]$$

Q5.23.1 MSQ 2023 1M Ans: A ● ○ ○

Given equation: $w^4 = 16j \Rightarrow w = (16j)^{1/4} = 2(0+j)^{1/4}$

Expanding using Euler's form: $w = 2\left[e^{j\frac{(2n+1)\pi}{2}}\right]^{1/4} = 2e^{j\frac{(2n+1)\pi}{8}}$ where $n = 0, 1, 2, 3$.

Evaluating the roots for verification:

- For $n = 0 \Rightarrow w = 2e^{j\pi/8}$
- For $n = 2 \Rightarrow w = 2e^{j5\pi/8}$
- For $n = 4 \Rightarrow w = 2e^{j9\pi/8}$ (Repeats periodically) $\boxed{\text{A. } 2e^{\frac{j2\pi}{8}}}$

Key Points

- **De Moivre's Theorem for Roots:** For a complex number $z = re^{j\theta}$, the n-th roots are given by:

$$z^{1/n} = r^{1/n}e^{j\left(\frac{\theta+2k\pi}{n}\right)} \quad \text{for } k = 0, 1, 2, \ldots, n-1$$

- **Geometry of Roots:** The n-th roots of a complex number always lie on a circle of radius $r^{1/n}$ centered at the origin, and they are equally spaced by an angle of $\frac{2\pi}{n}$ radians.

Q5.23.2 MCQ 2023 1M Ans: B ● ● ○

Let the contour integral be:

$$I = \oint_c \frac{z+2}{z^2+2z+2}\,dz \quad \text{where } c : \left|z + 1 - \frac{3}{2}i\right| = 1$$

Poles are found by setting the denominator to zero:

$$z^2 + 2z + 2 = 0 \Rightarrow (z+1)^2 + 1 = 0 \Rightarrow z + 1 = \pm\sqrt{-1}$$

$$\Rightarrow z = -1 + j, \quad -1 - j$$

The contour c is a circle centered at $\left(-1, \frac{3}{2}\right)$ with radius $R = 1$. Check which poles lie inside c:

- For $z = -1 + j \Rightarrow |(-1+j) + 1 - 1.5j| = |-0.5j| = 0.5 < 1$ (Inside c)
- For $z = -1 - j \Rightarrow |(-1-j) + 1 - 1.5j| = |-2.5j| = 2.5 > 1$ (Outside c)

Thus, only $z = -1 + j$ lies inside c, while $z = -1 - j$ lies outside.

By Cauchy's Residue Theorem:

$$\oint_c f(z)\,dz = 2\pi i\,\text{Res}(f(z), z = -1 + j)$$

$$= 2\pi i\left[\frac{z+2}{\frac{d}{dz}(z^2+2z+2)}\right]_{z=-1+j} = 2\pi i\left(\frac{z+2}{2(z+1)}\right)_{z=-1+j}$$

$$= 2\pi i\left(\frac{-1+j+2}{2(-1+j+1)}\right) = 2\pi i\left(\frac{1+j}{2j}\right) = \pi(1+j)$$

$$\boxed{\text{B. } \pi(1+j)}$$

Key Points

- Test poles algebraically against the circle equation $|z - z_0| = R$ by substituting z into the expression. If $|z - z_0| < R$, then z is inside the contour else no.
- **Simple Pole Evaluation:** For an isolated simple pole at $z = a$, the residue is straightforwardly calculated using $\text{Res}(f, a) = \lim_{z\to a}(z-a)f(z)$.

Q5.22.1 | NAT | 2022 | 1M | Ans: 0.5 | ● ● ○

In the given question,

$$f(z) = \frac{2^z}{z^2-1} = \frac{2^z}{(z-1)(z+1)}$$

The singularity points where $f(z)$ is not analytic lies at $z = 1$ and $z = -1$.
Since, $z = 1$ doesn't lie inside the contour C.
Hence,

$$I = \oint_c \frac{2^z}{z^2-1}\,dz = \oint_c \frac{\left(\frac{2^z}{z-1}\right)}{z+1}\,dz$$

$$I = 2\pi i\,[\text{Residue at } z = -1]$$

$$I = 2\pi i \times \lim_{z\to(-1)} (z+1)\cdot\frac{2^z}{(z-1)(z+1)}$$

$$I = 2\pi i \times \frac{2^{-1}}{(-1-1)} = -\frac{2\pi i}{4}$$

$$\therefore\ I = -\frac{\pi i}{2}$$

Given: $I = -\pi i A$. Hence,

$$A = \frac{1}{2} = 0.5$$

$$\boxed{0.50}$$

Q5.21.1 | MCQ | 2021 | 2M | Ans: MTA | ● ● ●

The given integral is

$$\oint_c \frac{\sin x}{x^2(x^2+4)}\,dx, \quad c : |x-i| = 2$$

Poles are given by

$$x^2 = 0 \quad \text{and} \quad x^2+4 = 0$$

Poles are given is

$$\Downarrow$$

$$\Rightarrow x = 0 \text{ is a pole of order '2'}$$

$$x = \pm 2i \text{ are simple nodes}$$

$$x = 0 \text{ lies inside 'c'}$$

$$x = 2i \text{ lies inside 'c'}$$

$$x = -2i \text{ lies outside 'c'}$$

Finding the residue by the formula we have.
Residue at $x = 0$

$$= \frac{1}{(2-1)!}\lim_{x\to 0}\frac{d}{dx}\left[(x-0)^2\frac{\sin x}{x^2(x^2+4)}\right]$$

$$= \lim_{x\to 0}\frac{(x^2+4)\cos x - \sin x(2x)}{(x^2+4)^2} = \frac{1}{4}$$

Residue at $x = 2i$

$$= \lim_{x\to 2i}(x-2i)\frac{\sin x}{x^2(x-2i)(x+2i)} = \frac{\sin(2i)}{(-4)(4i)}$$

By CRT

$$\oint_c f\,dx = 2\pi i\,[\text{Res}_0 + \text{Res}_{2i}] = 2\pi i\left[\frac{1}{4} - \frac{\sin(2i)}{16i}\right]$$

None of the option is correct.

$$\boxed{4.26}$$

Q5.19.1 MCQ 2019 1M Ans: D ● ● ○

Examine the analyticity and singularities for each of the given options:

A. $f(z) = \ln(z)$
The complex logarithmic function $\ln(z) = \ln|z| + j\arg(z)$ is multi-valued. To make it a single-valued analytic function, a branch cut is introduced (typically along the negative real axis from 0 to $-\infty$). The point $z = 0$ is a branch point singularity. Therefore, $\ln(z)$ is not analytic at $z = 0$ and along its branch cut.

B. $f(z) = e^{1/z}$
Laurent series expansion:

$$e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \dots$$

Since the principal part (the terms with negative powers of z) contains infinitely many non-zero terms, $z = 0$ is an **essential singularity**. The function is analytic everywhere except at $z = 0$.

C. $f(z) = \frac{1}{1-z}$
The function is a rational function. Its singularity occurs where the denominator equals zero:

$$1 - z = 0 \Rightarrow z = 1$$

Since it is a non-isolated zero of the denominator of order 1, $z = 1$ is a **simple pole**. The function is analytic everywhere except at $z = 1$.

D. $f(z) = \cos(z)$
Taylor series expansion:

$$\cos(z) = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots$$

The series converges for all $z \in \mathbb{C}$ ($R = \infty$). It contains no negative powers of z, meaning it possesses no singularities in the finite complex plane. Thus, $\cos(z)$ is an **entire function** (always analytic).

$$\boxed{D.\cos(z)}$$

Key Points

- **Classification of Singularities:**
  - **Pole:** If the principal part of the Laurent series has a finite number of terms.
  - **Essential Singularity:** If the principal part has infinitely many terms (e.g., $e^{1/z}$ at $z = 0$).
  - **Branch Point:** A point where a multi-valued function cannot be defined continuously as a single-valued function (e.g., $z = 0$ for $\ln z$).

Q5.19.2 NAT 2019 1M Ans: -0.0001 to 0.0001 ● ○ ○

$$\frac{1}{2\pi i}\oint_C \frac{(z^2+1)^2}{z^2}\,dz = \text{Res}\left[z^2 f(z)\right]$$

$$= \lim_{z\to 0} \frac{1}{1!}\frac{d}{dz} z^2 \times \frac{(z^2+1)^2}{z^2} = \lim_{z\to 0} 2\times 2z(z^2+1) = 0$$

$$\boxed{0}$$

Q5.18.1 MCQ 2018 2M Ans: 2 ● ● ○

Given integral:

$$\frac{1}{\pi j}\oint_O \frac{dz}{z^2-1} = \frac{1}{\pi j}\oint_O f(z)\,dz$$

The function is:

$$f(z) = \frac{1}{z^2-1}$$

It has 2 poles at $z = +1$ & $z = -1$. Based on contour shown, both lie inside contour but both are in opposite direction one clockwise and one anti clockwise.

$$f_{(z)} = \left[\frac{1}{(z-1)} - \frac{1}{z+1}\right] \times \frac{1}{2} = 2$$

$$\boxed{2}$$

Key Points

- **Contour Orientation Effects:** A positive (counterclockwise) orientation adds a positive residue contribution ($+2\pi i$ Res), whereas a negative (clockwise) orientation flips the sign of that specific pole's integration path contribution ($-2\pi i$ Res).

Q5.17.1 MCQ 2017 1M Ans: B ● ○ ○

Given function:

$$f(z) = \frac{1}{(z-4)(z+1)^3}$$

There are two poles, a simple pole at $z = 4$ and pole of order 3 at $z = -1$.

Residue at $z = 4$:

$$\text{Res} = \lim_{z\to 4}(z-4)f(z) = \lim_{z\to 4}\frac{1}{(z+1)^3} = \frac{1}{(4+1)^3} = \frac{1}{125}$$

Residue at $z = -1$:

$$\text{Res} = \lim_{z\to -1}\frac{1}{2!}\frac{d^2}{dz^2}\left[(z+1)^3 \times \frac{1}{(z-4)(z+1)^3}\right]$$

$$= \lim_{z\to -1}\frac{1}{2}\left[\frac{2}{(z-4)^3}\right] = \frac{1}{(-1-4)^3} = \frac{-1}{125}$$

$$\boxed{\text{B. } \frac{1}{125} \text{ and } \frac{-1}{125}}$$

Mistakes to be avoided

Do not drop the $\frac{1}{(m-1)!}$ coefficient before evaluating the final limits.

Q5.17.2 MCQ 2017 1M Ans: D ● ● ○

Given integral:

$$I = \oint_c \frac{z^2-1}{z^2+1}e^z\,dz$$

The poles are at $z = \pm j$. Both these poles lie inside given curve $|z| = 3$.

Residue at $z = j$:

$$\text{Res}(z=j) = \lim_{z\to j}(z-j)\frac{(z^2-1)}{(z^2+1)}e^z = \lim_{z\to j}\frac{(z^2-1)}{(z+j)}e^z = \frac{-2e^j}{2j} = je^j$$

Residue at $z = -j$:

$$\text{Res}(z=-j) = \lim_{z\to -j}(z+j)\frac{(z^2-1)}{(z^2+1)}e^z = \lim_{z\to -j}\frac{(z^2-1)}{(z-j)}e^z = \frac{-2e^{-j}}{-2j} = -je^{-j}$$

By Cauchy's Residue Theorem:

$$I = 2\pi j\left[je^j - je^{-j}\right] = 2\pi j \times j\,[2j\sin 1] = -4\pi j\sin 1$$

$$\boxed{\text{D. } -4\pi i\sin(1)}$$

Key Points

- **Euler's Identity Conversion:**

$$\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j} \implies e^{j\theta}-e^{-j\theta} = 2j\sin\theta$$

Q5.16.1 NAT 2016 2M Ans: -136: -132 ● ● ●

Let the given contour integral be:

$$I = -\frac{1}{2\pi}\oint_c \frac{\sin z}{(z-j2\pi)^3}\,dz$$

By generalized Cauchy Integral Formula:

$$\oint_c \frac{f(z)}{(z-z_0)^{n+1}}\,dz = 2\pi j\left\{\frac{f^{(n)}(z_0)}{n!}\right\}$$

- $f(z) = \sin z$ (analytic everywhere)
- $z_0 = j2\pi$ (assumed to lie inside the contour c)
- $n+1 = 3 \implies n = 2$

$$f'(z) = \cos z$$
$$f''(z) = -\sin z$$

At $z_0 = j2\pi$: Using the hyperbolic relation $\sin(j\theta) = j\sinh\theta$:

$$f''(j2\pi) = -\sin(j2\pi) = -j\sinh(2\pi)$$

$$\oint_c \frac{\sin z}{(z-j2\pi)^3}\,dz = 2\pi j \times \frac{f''(j2\pi)}{2!}$$
$$= 2\pi j \times \frac{-j\sinh(2\pi)}{2}$$
$$= \pi\sinh(2\pi)$$
$$I = -\frac{1}{2\pi}\times[\pi\sinh(2\pi)]$$
$$I = -\frac{1}{2}\sinh(2\pi)$$

Using the numerical value $\sinh(2\pi) \approx 267.74676$:

$$I = -\frac{1}{2}\times 267.74676 = -133.87338$$

$$\boxed{-133.87}$$

Key Points

- **Generalized Cauchy Integral Formula:** High-order poles in denominators dictate matching derivative iterations of the numerator mapped directly to the singular coordinate point:

$$\oint_c \frac{f(z)}{(z-z_0)^{n+1}}dz = \frac{2\pi j}{n!}f^{(n)}(z_0)$$

- **Circular to Hyperbolic Identities:**

$$\sin(j\theta) = j\sinh\theta \quad \text{and} \quad \cos(j\theta) = \cosh\theta$$

Q5.16.2 NAT 2016 1M Ans: 0 ● ● ●

Let $z = x + iy$. The given complex function is:

$$f(z) = 2(x+iy)^3 + b|x+iy|^3$$

Expanding the cubic term and using the modulus definition $|z| = \sqrt{x^2+y^2}$:

$$f(z) = 2\left\{x^3 + 3x^2(iy) - 3xy^2 - iy^3\right\} + b\left(x^2+y^2\right)^{3/2}$$

$$f(z) = 2(x^3 - 3xy^2) + b\left(x^2+y^2\right)^{3/2} + i(6x^2y - 2y^3)$$

Let $f(z) = u + iv$, where the real part u and imaginary part v are:

$$u = 2(x^3 - 3xy^2) + b\left(x^2+y^2\right)^{3/2}$$

$$v = 6x^2y - 2y^3$$

Now, we evaluate the partial derivatives required for the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = 6x^2 - 6y^2 + 3bx\left(x^2+y^2\right)^{1/2}$$

$$\frac{\partial v}{\partial y} = 6x^2 - 6y^2$$

For $f(z)$ to be analytic everywhere, it must satisfy the first Cauchy-Riemann condition ($\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$) identically for all points (x, y):

$$6x^2 - 6y^2 + 3bx\left(x^2+y^2\right)^{1/2} = 6x^2 - 6y^2$$

$$\Rightarrow 3bx\left(x^2+y^2\right)^{1/2} = 0$$

Since this must hold true for all values of x and y, the coefficient must vanish:

$$\therefore b = 0$$

$$\boxed{0}$$

Key Points

- **Cauchy-Riemann Equations:** A complex function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain if its partial derivatives are continuous and satisfy:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Mistakes to be avoided

- Do not treat functions containing $|z|$, $\bar{z}$, or $\text{Re}(z)$ as automatically differentiable with respect to z.

Q5.16.3 MCQ 2016 2M Ans: B ● ● ○

Let the given complex integral expression be:

$$I = \frac{1}{2\pi j} \oint_C \frac{e^z}{z-2}\, dz$$

Simple pole:

$$z - 2 = 0 \implies z = 2$$

Case 1: If $z = 2$ lies inside the contour C By Cauchy's Integral Formula / Cauchy's Residue Theorem

$$\oint_C \frac{g(z)}{z - z_0}\, dz = 2\pi j \cdot g(z_0)$$

Here, $g(z) = e^z$ and $z_0 = 2$. Therefore:

$$I = \frac{1}{2\pi j} \times 2\pi j\, [\text{Res}(f(z), z = 2)]$$

$$I = \frac{1}{2\pi j} \times 2\pi j \cdot e^2 = e^2$$

Since $e \approx 2.71828$:

$$I = (2.71828)^2 \approx 7.39$$

Case 2: If $z = 2$ lies outside the contour C By Cauchy's Goursat Theorem, the closed line integral of any function over an analytic domain is identically zero:

$$I = \frac{1}{2\pi j} \oint_C \frac{e^z}{z-2}\, dz = 0$$

$$\boxed{\text{B. (i) 7.39 (ii) 0}}$$

Q5.15.1 MCQ 2015 1M Ans: D ● ● ○

(a) Residue of $f(z) = \frac{z}{z^2-1}$ at $z = 1$ is $\frac{1}{2}$:

$$f(z) = \frac{z}{(z-1)(z+1)}$$

The point $z = 1$ is a simple pole. Calculating its residue:

$$\text{Res}(f, 1) = \lim_{z\to 1} (z-1)\frac{z}{(z-1)(z+1)} = \lim_{z\to 1} \frac{z}{z+1} = \frac{1}{1+1} = \frac{1}{2}$$

Thus, statement (a) is correct.

(b) $\oint_C z^2\, dz = 0$ The function $g(z) = z^2$ is a polynomial function, which means it is an entire function (analytic everywhere in the finite complex plane).

$$\oint_C z^2\, dz = 0$$

Thus, statement (b) is correct.

(c) $\frac{1}{2\pi i}\oint_C \frac{dz}{z} = 1$ The integrand has a simple pole at $z = 0$. By Cauchy's Integral Formula,

$$\oint_C \frac{1}{z}\,dz = 2\pi i \cdot (1) \implies \frac{1}{2\pi i}\oint_C \frac{dz}{z} = 1$$

Thus, statement (c) is correct.

(d) $f(z) = \bar{z}$ **is an analytic function:** Let $z = x + iy$, so complex conjugate $\bar{z} = x - iy$. $f(z) = u(x, y) + iv(x, y)$, we isolate the real part u and imaginary part v:

$$u = x \quad \text{and} \quad v = -y$$

Check the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = 1, \quad \frac{\partial v}{\partial y} = -1 \implies \frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$$

Since the Cauchy-Riemann criteria are violated everywhere, the function $f(z) = \bar{z}$ is nowhere differentiable and not analytic. Therefore, statement (d) is false.

$$\boxed{\text{D. } \bar{z} \text{ (complex conjugate of } z\text{) is an analytical function.}}$$

Q5.15.2 MCQ 2015 1M Ans: B ● ○ ○

Given matrix A:

$$A = \begin{bmatrix} 10 & 5+j & 4 \\ x & 20 & 2 \\ 4 & 2 & -10 \end{bmatrix}$$

Method 1:Using Hermitian Matrix Properties

All eigenvalues of a Hermitian matrix are strictly real. A square matrix is Hermitian if it equals its own conjugate transpose ($A = A^H$):

$$A = (\bar{A})^T$$

Taking the conjugate transpose of A:

$$A^H = \begin{bmatrix} 10 & \bar{x} & 4 \\ \overline{5+j} & 20 & 2 \\ 4 & 2 & -10 \end{bmatrix} = \begin{bmatrix} 10 & \bar{x} & 4 \\ 5-j & 20 & 2 \\ 4 & 2 & -10 \end{bmatrix}$$

For the eigenvalues of A to be guaranteed real, $A = A^H$.

$$\boxed{x = 5 - j}$$

Method 2: Characteristic Polynomial

Alternatively, we can find the eigenvalues by solving the characteristic equation $\det(A - \lambda I) = 0$:

$$\det \begin{bmatrix} 10-\lambda & 5+j & 4 \\ x & 20-\lambda & 2 \\ 4 & 2 & -10-\lambda \end{bmatrix} = 0$$

Expanding the determinant along the first row:

$$(10-\lambda)\begin{vmatrix} 20-\lambda & 2 \\ 2 & -10-\lambda \end{vmatrix} - (5+j)\begin{vmatrix} x & 2 \\ 4 & -10-\lambda \end{vmatrix} + 4\begin{vmatrix} x & 20-\lambda \\ 4 & 2 \end{vmatrix} = 0$$

Evaluating each minor determinant gives:

$$\begin{aligned}(10-\lambda)\left[(20-\lambda)(-10-\lambda)-4\right] &= -\lambda^3+20\lambda^2-104\lambda-2040\\ -(5+j)\left[x(-10-\lambda)-8\right] &= (5+j)x\lambda+10(5+j)x+8(5+j)\\ +4\left[2x-4(20-\lambda)\right] &= 8x-320+16\lambda\end{aligned}$$

We obtain:

$$-\lambda^3+20\lambda^2+\lambda\left[-88+(5+j)x\right]+\left[-2320+8j+(58+10j)x\right]=0$$

For all roots (λ) of this cubic equation to be purely real numbers, every single coefficient in the characteristic polynomial must be completely real. Let $x = a + jb$

$$\text{Im}\left\{-88+(5+j)(a+jb)\right\}=0 \implies a+5b=0 \implies a=-5b \quad \text{— (Eq. 1)}$$

$$\text{Im}\left\{-2320+8j+(58+10j)(a+jb)\right\}=0 \implies 8+10a+58b=0 \quad \text{— (Eq. 2)}$$

Substituting Eq. 1 into Eq. 2:

$$8+10(-5b)+58b=0 \implies 8+8b=0 \implies b=-1$$

Putting $b = -1$ back into Eq. 1 gives $a = 5$. Thus, combining our components:

$$\boxed{x=5-j}$$

$$\boxed{B.\,5-j}$$

Q5.15.3 NAT 2015 1M Ans: 9.9 to 10.1 ● ● ○

Given the Möbius (bilinear) transformation:

$$f(z)=\frac{az+b}{cz+d}$$

Given that $f(z_1) = f(z_2)$ for all $z_1 \neq z_2$. This implies that the function maps distinct points to the same value, hence a constant function. The numerator must be a constant multiple of the denominator,therefore the determinant of the transformation matrix must be zero:

$$ad-bc=0$$

Given the values: $a = 2$, $b = 4$, $c = 5$ Substituting these values into the condition:

$$2\cdot d-4\cdot 5=0$$

$$2d-20=0$$

$$2d=20$$

$$d=10$$

Thus, the value of d must be equal to 10.

$$\boxed{10}$$

Key Points

- If $ad - bc \neq 0$, the transformation is bijective (one-to-one) on the extended complex plane and ensures transformation is non-degenerate.

Q5.15.4 NAT 2015 2M Ans: 0.5 ● ● ○

$$I = \frac{1}{2\pi j} \oint_C \text{Re}(z)\, dz$$

where the contour C is the unit circle given by $|z| = 1$.

$$z = e^{j\theta} = \cos\theta + j\sin\theta \quad \text{for } \theta \in [0, 2\pi]$$

The differential dz is given by:

$$dz = je^{j\theta}\, d\theta$$

The real part of z on this contour is:

$$\text{Re}(z) = \cos\theta$$

Substituting these expressions into the integral I:

$$I = \frac{1}{2\pi j} \int_0^{2\pi} \cos\theta \cdot \left(je^{j\theta}\right) d\theta$$

$$I = \frac{1}{2\pi} \int_0^{2\pi} \cos\theta \cdot (\cos\theta + j\sin\theta)\, d\theta$$

$$I = \frac{1}{2\pi} \left\{ \int_0^{2\pi} \cos^2\theta\, d\theta + j \int_0^{2\pi} \cos\theta \sin\theta\, d\theta \right\}$$

$$\int_0^{2\pi} \cos^2\theta\, d\theta = \int_0^{2\pi} \frac{1 + \cos(2\theta)}{2}\, d\theta = \pi$$

$$\int_0^{2\pi} \cos\theta \sin\theta\, d\theta = \int_0^{2\pi} \frac{\sin(2\theta)}{2}\, d\theta = 0$$

Substituting these values back into the equation for I:

$$I = \frac{1}{2\pi}(\pi + j \cdot 0) = \frac{\pi}{2\pi} = \frac{1}{2} = 0.5$$

$$\boxed{0.5}$$

Key Points

- Parameterization of a unit circle contour: $z = e^{j\theta}$, $dz = je^{j\theta}\, d\theta$ where $0 \le \theta \le 2\pi$.
- Non-analytic functions like $\text{Re}(z)$ require explicit path integration as Cauchy's Integral Theorem does not apply.

Q5.15.5 MCQ 2015 1M Ans: B ● ● ○

According to Cauchy's Integral Formula for higher-order derivatives:

$$\oint_C \frac{f(z)\, dz}{(z - z_0)^{n+1}} = 2\pi i \left\{ \frac{f^n(z_0)}{n!} \right\}$$

Given,

$$f(z) = 1$$

Since $f(z)$ is a constant function, its n-th derivative for any integer $n \ge 1$ is zero:

$$f^n(z_0) = 0$$

Substituting this value back into the formula yields:

$$\oint_C \frac{dz}{(z - z_0)^{n+1}} = 0$$

$$\boxed{B.\ 0}$$

Mistakes to be avoided

- Ensure that $n \geq 1$; if $n = 0$, the integral evaluates to $2\pi i \cdot f(z_0) = 2\pi i$.

Q5.14.1 MCQ 2014 2M Ans: B ● ● ●

An analytic function $f(z)$ can be expressed in terms of its real and imaginary parts as:

$$f(z) = u + iv$$

Given the real part:

$$u = e^{-y}\cos x$$

Differentiating u partially with respect to x:

$$u_x = -e^{-y}\sin x$$

By the Cauchy-Riemann equations, we know that:

$$u_x = v_y$$

Therefore:

$$v_y = -e^{-y}\sin x$$

Integrating both sides with respect to y:

$$v = e^{-y}\sin x + \phi(x) \quad \text{— (i)}$$

Now, differentiating u partially with respect to y:

$$u_y = -e^{-y}\cos x$$

Differentiating equation (i) partially with respect to x:

$$v_x = e^{-y}\cos x + \phi'(x)$$

By the Cauchy-Riemann equations, we also have:

$$u_y = -v_x$$

Substituting the partial derivatives into this relation:

$$-e^{-y}\cos x = -\left(e^{-y}\cos x + \phi'(x)\right)$$

$$-e^{-y}\cos x = -e^{-y}\cos x - \phi'(x)$$

This simplifies to:

$$\phi'(x) = 0$$

Integrating with respect to x gives:

$$\phi(x) = C \quad (\text{where } C \text{ is a constant})$$

Substituting $\phi(x)$ back into equation (i) yields the imaginary part v:

$$v = e^{-y}\sin x + C$$

$$\boxed{B.\ e^{-y}\sin(x)}$$

Key Points

If the real part u is known, the conjugate harmonic function v can be uniquely determined up to an additive constant using these relations.

Q5.14.2 MCQ 2014 1M Ans: B ● ● ●

An all-pass system has a constant magnitude response of unity across all frequencies, meaning $|H(e^{j\omega})| = 1$. Given the transfer function of the first-order all-pass filter:

$$H(z) = \frac{z^{-1} - b}{1 - az^{-1}}$$

Substituting $z = e^{j\omega}$ to evaluate the frequency response on the unit circle:

$$H(e^{j\omega}) = \frac{e^{-j\omega} - b}{1 - ae^{-j\omega}}$$

Taking the squared magnitude, we set $|H(e^{j\omega})|^2 = 1$:

$$\left|\frac{e^{-j\omega} - b}{1 - ae^{-j\omega}}\right|^2 = 1 \implies |e^{-j\omega} - b|^2 = |1 - ae^{-j\omega}|^2$$

Expanding both sides using the identity $|x|^2 = x \cdot x^*$:

$$(e^{-j\omega} - b)(e^{j\omega} - b^*) = (1 - ae^{-j\omega})(1 - a^*e^{j\omega})$$

$$1 - b^*e^{-j\omega} - be^{j\omega} + |b|^2 = 1 - ae^{-j\omega} - a^*e^{j\omega} + |a|^2$$

Comparing Coefficients Since this equality must hold for all values of ω, the coefficients of the corresponding exponential terms on both sides must be equal:
Coefficient of $e^{j\omega}$:

$$-b = -a^* \implies b = a^*$$

Coefficient of $e^{-j\omega}$:

$$-b^* = -a \implies b = a^*$$

Constant terms:

$$1 + |b|^2 = 1 + |a|^2 \implies |b|^2 = |a|^2$$

(which holds true since $|a^*|^2 = |a|^2$) Thus, the parameter b must be equal to the complex conjugate of a (a^*).

$$\boxed{B.\ a^*}$$

Key Points

For all-pass systems poles must lie on the left and zeroes on the mirror image of the pole that is on the right. If pole is at $z = p$, then zero will be at $z = 1/p^*$.

Q5.14.3 NAT 2014 2M Ans: -0.6 to -0.4 ● ● ●

Given:

$$H_1(z) = (1 - pz^{-1})^{-1}$$

$$H_2(z) = (1 - qz^{-1})^{-1}$$

$$H(z) = H_1(z) + rH_2(z)$$

$$p = \frac{1}{2},\ q = -\frac{1}{4},\ |r| < 1$$

and zero of H(z) lies on the unit circle.

$$H(z) = \frac{1}{1 - pz^{-1}} + \frac{r}{1 - qz^{-1}}$$

$$H(z) = \frac{(1 - qz^{-1}) + r(1 - pz^{-1})}{(1 - pz^{-1})(1 - qz^{-1})}$$

$$H(z) = \frac{1 - qz^{-1} + r - prh^{-1}}{(1 - pz^{-1})(1 - qz^{-1})}$$

$$H(z) = \frac{((1 + r) - (pr + q)z^{-1})}{(1 - pz^{-1})(1 - qz^{-1})}$$

Zero of H(z) $\Rightarrow (1 + r) - (pr + q)z^{-1} = 0$

$$(1 + r) = (pr + q)z^{-1}$$

$$z = \frac{pr + q}{1 + r} = \frac{\frac{1}{2}r - \frac{1}{4}}{1 + r}$$

Given $|z| = 1$ [Zero lies on unit circle]

$$z = \pm 1$$

Taking $z = 1$

$$1 = \frac{\frac{1}{2}r - \frac{1}{4}}{1 + r}$$

$$r = -\frac{5}{2}$$

Taking $z = -1$

$$-1 = \frac{\frac{1}{2}r - \frac{1}{4}}{1 + r}$$

Solving the above we get,

$$r = -0.5$$

And Given that $|r| < 1$ Hence $r = -\frac{5}{2}$ cannot be possible

$$\therefore r = -\frac{1}{2} \text{ ; } |r| < 1 \text{ is satisfied.}$$

$$\boxed{-0.5}$$

Q5.12.1 MCQ | 2012 | 1M | Ans: C | ● ○ ○

$$f(z) = \frac{1}{z + 1} - \frac{2}{z + 3}$$

$$\frac{1}{2\pi j} \oint_C f(z)\, dz = \frac{1}{2\pi j} \oint_C \frac{dz}{z + 1} - \frac{1}{2\pi j} \oint_C \frac{2dz}{z + 3}$$

C : $|z + 1| = 1$ $z = -3$ which lies outside $z = -1$ lies inside So by Cauchy's Residue Theorem

$$\because \int_C f(z)\, dz = 2\pi j \left(\sum_{n=1}^{n} R_n \right) \quad \text{where } R_n = \text{Res}\left[f(z); z = z_n \right]$$

$$\frac{1}{2\pi j} \int_C f(z)\, dz = \frac{1}{2\pi j} \times 2\pi j(1) - 0 = 1$$

$$\boxed{C.\ 1}$$

Q5.11.1 MCQ 2011 1M Ans: A ● ● ○

$$I = \oint_C \frac{-3z+4}{z^2+4z+5} dz \; ; \; |z| = 1$$

$$z^2 + 4z + 5$$

$z = -2 + i$ & $-2 - i$ are poles which lie outside $|z| = 1$ So by Cauchy's Residue Theorem

$$I = 0$$

$$\boxed{A.\, 0}$$

Q5.10.1 MCQ 2010 2M Ans: C ● ○ ○

$$\mathrm{x}(z) = \frac{1-2z}{z(z-1)(z-2)}$$

$z = 0, 1, 2$ are poles

$$\mathrm{Res}\left[\mathrm{x}(z); 0\right] = \lim_{z\to 0} z\frac{(1-2z)}{z(z-1)(z-2)} = \frac{1}{(-1)(-2)} = \frac{1}{2}$$

$$\mathrm{Res}\left[\mathrm{x}(z); 1\right] = \lim_{z\to 1} \frac{(z-1)(1-2z)}{z(z-1)(z-2)} = \frac{(-1)}{1(-1)} = 1$$

$$\mathrm{Res}\left[\mathrm{x}(z); 2\right] = \lim_{z\to 2} \frac{(z-2)(1-2z)}{z(z-1)(z-2)}$$

$$= \frac{1-4}{2(2-1)} = -\frac{3}{2}$$

$$\boxed{C.\, \frac{1}{2}, 1 \text{ and } -\frac{3}{2}}$$

Q5.09.1 MCQ 2009 1M Ans: D ● ● ○

$$f(z) = C_0 + C_1 z^{-1} = \frac{C_1 + C_0 z}{z}$$

$$I = \oint \frac{1 + f(z)}{z} dz = \oint_{\substack{\text{unit}\\ \text{circle}}} \frac{C_1 + (1 + C_0)z}{z^2} dz$$

$$\because \mathrm{Res}\{f(z), z = z_0\} = \frac{1}{(m-1)!} \lim_{z\to z_0} \left[\frac{d^{m-1}}{dz^{m-1}} (z - z_0)^m f(z)\right]$$

Where m = order of the pole $z = 0$ is a pole order 2

$$I = 2\pi j \frac{f'(0)}{1!}$$

$$f(z) = C_1 + (1 + C_0)z$$

$$f'(z) = 1 + C_0$$

$$I = 2\pi j(1 + C_0)$$

$$\boxed{D.\, 2\pi j(1 + c_0)}$$

Q5.08.1 MCQ 2008 1M Ans: D ● ● ●

$$\sin(z) = 10$$
$$\frac{e^{jz} - e^{-jz}}{2j} = 10$$
$$e^{2jz} - j20e^{jz} - 1 = 0$$
$$e^{jz} = \frac{j20 \pm \sqrt{-400+4}}{2} = \frac{j20 \pm 6j\sqrt{11}}{2} = 10j \pm 3j\sqrt{11}$$
$$jz = \ln\left[10j \pm 3j\sqrt{11}\right] = \ln j + \ln\left[10 \pm 3\sqrt{11}\right]$$
$$jz = \ln e^{j\frac{\pi}{2} \pm 2n\pi} + \ln\left[10 \pm 3\sqrt{11}\right]$$
$$jz = j\left(\frac{\pi}{2} \pm 2n\pi\right) + \ln\left[10 \pm 3\sqrt{11}\right]$$

Hence this equation has infinite number of complex solutions.

$$\boxed{D.\text{an infinite number of complex solutions}}$$

Q5.08.2 MCQ 2008 2M Ans: A ● ○ ○

$$\because \operatorname{Res}\{f(z), z = z_0\} = \frac{1}{(m-1)!}\lim_{z\to z_0}\left[\frac{d^{m-1}}{dz^{m-1}}(z-z_0)^m f(z)\right]$$

Where m = order of the pole As z = 2 is a pole of order 2

$$\operatorname{Res}\{f(z), z = 2\} = \frac{f'(2)}{2!} \ldots\ldots\ldots\ldots\ldots (i)$$

Where

$$f(z) = \frac{1}{(z+2)^2}$$
$$f'(z) = \frac{-2}{(z+2)^3}$$
$$f'(2) = \frac{-2}{64} = -\frac{1}{32}$$
$$\operatorname{Res}\{f(z), 2\} = -\frac{1}{32}$$
$$\boxed{A. -\frac{1}{32}}$$

Q5.07.1 MCQ 2007 2M Ans: A ● ○ ○

$$I = \oint \frac{ds}{s^2-1} = \oint \frac{ds}{(s-1)(s+1)} = 2\pi j\ [\text{Sum of residues}]$$

From the figure given , $s = -1$ is outside D and $s = 1$ is inside

$$\{f(s), 1\} = \frac{1}{1+1} = \frac{1}{2}$$
$$\oint \frac{ds}{s^2-1} = 2\pi j\left\{\frac{1}{2}\right\} = \pi j$$
$$\boxed{A. j\pi}$$

Q5.06.1 MCQ 2006 2M Ans: B ● ○ ○

$$W = \ln Z = \ln(x + jy) = \ln\left[\sqrt{x^2 + y^2}\mathrm{e}^{\mathrm{j}\tan^{-1}\left(\frac{y}{x}\right)}\right]$$

$$W = \frac{1}{2}\ln(x^2 + y^2) + \mathrm{i}\tan^{-1}\frac{y}{x}$$

$$u = \frac{1}{2}\ln(x^2 + y^2)$$

As u = constant

$$x^2 + y^2 = \mathrm{e}^{2u}$$

Hence it is a set of concentric circles.

$$\boxed{B.\text{Set of concentric circles}}$$

Q5.06.2 MCQ 2006 2M Ans: D ● ○ ○

$$\oint_{|z-j|=2} \frac{1}{z^2 + 4}\,dz = \oint_{|z-j|=2} \frac{1}{(z-2j)(z+2j)}\,dz$$

Pole = 2j, -2j Only $z = 2j$ lies in the $|z - j| = 2$ region

$$\oint_{|z-j|=2} \frac{1}{z^2 + 4}\,dz = 2\pi j \text{ [Sum of residues]}$$

$$\oint_{|z-j|=2} \frac{1}{z^2 + 4}\,dz = 2\pi j \times \frac{1}{2j + 2j} = \frac{\pi}{2}$$

$$\boxed{D.\pi/2}$$

Q5.26.1 NAT 2026 2M Ans: 25 to 25.6 ● ● ○

The given integral is:

$$\oint_C \frac{(z+1)^2}{(z-i)(z-2)}\,dz$$

The contour is given by the circle $C : |z-(2+i)| = \frac{3}{2}$. The singular points of the integrand are at $z = i$ and $z = 2$. To determine which singular points lie inside the circle, we find their distance from the center of the circle, which is at $2+i$, and compare it with the radius $R = \frac{3}{2}$. Distance between the center and $z = i$:

$$|(2+i)-i| = |2| = 2$$

Since $2 > \frac{3}{2}$, the point $z = i$ lies outside the contour C. Distance between the center and $z = 2$:

$$|(2+i)-2| = |i| = 1$$

Since $1 < \frac{3}{2}$, the point $z = 2$ lies inside the contour C. Since only $z = 2$ lies inside the contour, we use the Cauchy's Residue Theorem:

$$\oint_C f(z)\,dz = 2\pi i \times (\text{Residue at } z = 2)$$

The residue at the simple pole $z = 2$ is found by:

$$\text{Res at } z = 2 = \lim_{z\to 2}(z-2)\cdot\frac{(z+1)^2}{(z-i)(z-2)} = \frac{(2+1)^2}{(2-i)} = \frac{9}{2-i}$$

Thus, the value of the integral is:

$$\oint_C f(z)\,dz = 2\pi i \times \frac{9}{2-i}$$

The magnitude of this integral is given by:

$$\left|\oint_C f(z)\,dz\right| = |2\pi i| \times \frac{9}{|2-i|} = 2\pi \times \frac{9}{\sqrt{2^2+(-1)^2}} = \frac{18\pi}{\sqrt{5}} \approx 25.3$$

$$\boxed{25.3}$$

Q5.25.1 NAT 2025 2M Ans: 0 ● ● ○

Method 1

Given: $f(z) = jz$

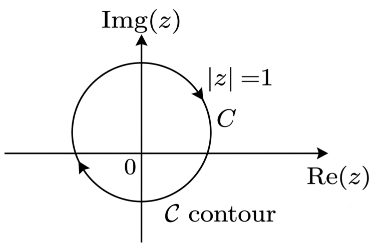

Fig. Solution Q5.25.1

Contour: $|z| = 1$

$$\therefore z = e^{j\theta}$$

$$dz = je^{j\theta}d\theta$$

$$I = \oint_C f(z)\,dz = -\int_0^{2\pi} je^{j\theta}\left(je^{j\theta}\right)d\theta$$

(-ve sign as contour is in clockwise direction, which is negative direction of angle)

$$\therefore I = \int_0^{2\pi} -e^{j2\theta} d\theta = \left[\frac{e^{j2\theta}}{j2}\right]_0^{2\pi}$$

$$= \frac{e^{j4\pi} - e^0}{j2} = 0$$

$$\therefore I = 0$$

Method 2

$f(z) = jz$. No poles of $f(z)$ exist inside the contour $|z| = 1 \therefore f(z)$ is an analytic function inside the contour $|z| = 1$. By Cauchy Integral Theorem

$$\therefore \oint_C f(z)\, dz = 0$$

$$\boxed{0}$$

Key Points

- **Cauchy Integral Theorem:** If a function $f(z)$ is analytic at all points interior to and on a simple closed contour C, then $\oint_C f(z)\, dz = 0$.
- Clockwise traversal introduces a negative sign relative to standard counter-clockwise parameters.

Q5.24.1 MCQ 2024 1M Ans: D ● ● ○

Analytic functions don't contain $|z|$, $\text{Re}(z)$ or $\text{Im}(z)$ post simplification.

$$f(z) = z^2 - z = (x + iy)^2 - (x + iy)$$

$$f(z) = (x^2 - y^2 - x) + i(2xy - y)$$

$$f(z) = u + iv$$

$$u = (x^2 - y^2 - x) \quad \text{and} \quad v = (2xy - y)$$

By Cauchy Riemann Equations

$$\frac{\partial u}{\partial x} = 2x - 1 \quad \text{and} \quad \frac{\partial u}{\partial y} = -2y$$

$$\frac{\partial v}{\partial x} = 2y \quad \text{and} \quad \frac{\partial v}{\partial y} = 2x - 1$$

Since, $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ Cauchy Riemann Equations are satisfied so function is analytic.

$$\boxed{D.\ f(z) = z^2 - z}$$

Key Points

- **Polynomial Analyticity:** Complex polynomial expressions involving only powers of z (and no $\bar{z}$, $|z|$, $\text{Re}(z)$, or $\text{Im}(z)$) are generally everywhere analytic (entire).

Q5.24.2 NAT 2024 1M Ans: 0 ● ● ○

Expanding the given function with the help of Taylor Series

$$f(z) = \cos z + e^{z^2}$$

$$= \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \ldots\infty\right) + \left(1 + \frac{z^2}{1!} + \frac{z^4}{2!} + \frac{z^6}{3!} + \ldots\infty\right)$$

$$\Rightarrow \text{no } z^5 \text{ term present}$$

$$\boxed{0}$$

Q5.21.1 MCQ 2021 1M Ans: D ● ○ ○

As the sum of the roots is a complex number. So it can be said that, At least one root is complex.So all the roots cannot be real.

$$\boxed{D.\text{All roots cannot be real.}}$$

Key Points

- **Properties of Polynomial Roots:** For any polynomial with coefficients, if the sum of the roots ($\sum \alpha_i = -a_{n-1}/a_n$) results in a complex number with a non-zero imaginary part, it is impossible for all individual roots to be purely real numbers.

Q5.21.2 MCQ 2021 2M Ans: C ● ● ○

Given: the contour integral $\oint_C \frac{dz}{z^2(z-4)}$

Singularities are given by $z^2(z-4) = 0 \implies z = 0, 4$

$z = 0$ is pole of order $m = 2$ lies inside contour 'c' and $z = 4$ is pole of order $m = 1$ lies outside 'c'

Residue at $x = 0$ is equal to

$$= \frac{1}{(2-1)!} \lim_{z\to 0} \frac{d^{2-1}}{dz^{2-1}}\left[(z-0)^2 \frac{1}{z^2(z-4)}\right]$$

$$= \frac{-1}{(0-4)^2} = \frac{-1}{16}$$

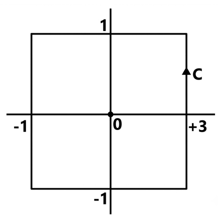

Fig. Solution Q5.21.2

By Cauchy residue theorem, we get

$$\oint_c f(z)\,dz = 2\pi i\,\text{Res} = 2\pi i\left[\frac{-1}{16}\right] = \frac{-\pi i}{8}$$

$$\boxed{C. -\frac{\pi j}{8}}$$

Q5.20.1 MCQ 2020 1M Ans: C ● ○ ○

Complex integral is given as $\int \frac{z^2+1}{z^2-2z}\,dz$ where C is $|z| = 1$

$$= \int \frac{z^2+1}{z(z-2)}\,dz = \int \frac{\frac{z^2+1}{z-2}}{z}\,dz$$

By integral formula, $2\pi i f(0)$ where, it solves to

$$f(z) = \frac{z^2+1}{z-2} = 2\pi i\left(\frac{1}{-2}\right) = -\pi i$$

$$\boxed{C. -\pi i}$$

Q5.19.1 MCQ 2019 1M Ans: B ● ○ ○

Given region $|z| \leqslant 1$

(a) $\frac{z^2-1}{z}$ $z = 0 \Rightarrow |z| = 0 \leqslant 1$ The pole is lies inside the given region. Hence, the function is not analytic.

(b) $\frac{z^2-1}{z+2}$ $z+2 = 0 \rightarrow z = -2 \Rightarrow |z| = 2 \geqslant 1$ the pole is lies outside the given region. Hence, the function is analytic.

(c) $\frac{z^2-1}{z-0.5}$ $z - 0.5 = 0 \Rightarrow z = 0.5 \Rightarrow |z| = 0.5 \leqslant 1$ The pole is lies inside the given region. Hence, the function is not analytic.

(d) $\frac{z^2-1}{z+j0.5}$ $z + j0.5 = 0 \Rightarrow z = -j0.5 \Rightarrow |z| = 0.5 \leqslant 1$ The pole is lies inside the given region. Hence, the function is not analytic.

$$\boxed{B.\frac{z^2-1}{z+2}}$$

Key Points

- **Analyticity of a Rational Function:** A rational function is analytic everywhere except at its poles (where the denominator equals zero).

Q5.19.2 MCQ 2019 2M Ans: A ● ○ ○

$$\oint_{|z|=5} \frac{z^3+z^2+8}{z+2}\,dz$$

The pole is at $z = -2$. Since $|-2| = 2 \leqslant 5$, the pole lies within the circle $|z| = 5$.

$$= 2\pi i \text{Res}\left[(z+2)f(z)\right]_{z=-2}$$

$$= 2\pi i \lim_{z\to-2}(z^3+z^2+8) = 8\pi i$$

$$\boxed{A. 8\pi j}$$

Q5.18.1 | MCQ | 2018 | 1M | Ans: A | ● ● ○

Given the integral:

$$\oint_C \frac{z+1}{z^2-4}\,dz$$

where the contour C is $|z+2| = 1$.
The poles of the integrand $f(z) = \frac{z+1}{z^2-4} = \frac{z+1}{(z+2)(z-2)}$ are located at $z = 2$ and $z = -2$.
pole $z = -2$: $|-2+2| = 0 < 1$ (lies inside the contour C).
pole $z = 2$: $|2+2| = 4 > 1$ (lies outside the contour C).
This contour only encircles $Z = -2$ pole & hence we need residue at $z = -2$:

$$\oint_C \frac{z+1}{z^2-4}\,dz = 2\pi i \text{Res}[f(z)]_{z=-2}$$

$$= 2\pi i \lim_{z\to -2}(z+2)\left[\frac{(z+1)}{(z+2)(z-2)}\right] = \frac{2\pi i}{4} = \frac{\pi i}{2}$$

$$\boxed{A.\frac{\pi i}{2}}$$

Q5.18.2 | MCQ | 2018 | 2M | Ans: B | ● ○ ○

$$\oint_c \frac{z^2}{\left(z^2-3z+2\right)^2}\,dz = \oint_c \frac{z^2}{(z-1)^2(z-2)^2}\,dz$$

Based on curve given both poles $z = 1$, $z = 2$; lie inside the curve, Hence, the integration over closed curve would be zero.

$$\boxed{B.0}$$

Q5.17.1 | MCQ | 2017 | 1M | Ans: D | ● ○ ○

$$\lim_{z\to i} \frac{z^2+1}{z^3+2z-i(z^2+2)}$$

As $z \to i$, the expression becomes

$$\frac{0}{0}$$

Applying L'Hospital rule,

$$\lim_{z\to i} \frac{2z}{(3z^2+2)-2iz} = \frac{2i}{-1+2} = 2i$$

$$\boxed{D.2i}$$

Q5.17.2 | MCQ | 2017 | 2M | Ans: B | ● ● ○

$$Z = x + iy$$

$$dz = (dx + idy)$$

The equation of line is

$$y = x$$

$$dz = (1+i)\,dx$$

$$I = \int_0^1 (1+i)x^2(1+i)\,dx$$

$$= (1+i)^2 \left[\frac{x^3}{3}\right]_0^1$$

$$I = (1+i)^2 \times \frac{1}{3} = \frac{2i}{3}$$

$$\boxed{B. \frac{2i}{3}}$$

Q5.16.1 MCQ | 2016 | 1M | Ans: B | ● ● ○

$$I = \oint_C \frac{2z+5}{\left(z-\frac{1}{2}\right)\left(z^2-4z+5\right)} dz$$

Pole

$$z = \frac{1}{2}$$

lies inside circle

$$|z| = 1.$$

$$I = 2\pi i \times \left\{\text{Residue of } z \text{ at } z = \frac{1}{2}\right\}$$

$$I = 2\pi i \frac{\left\{2 \times \frac{1}{2} + 5\right\}}{\left\{\left(\frac{1}{2}\right)^2 - 4 \times \frac{1}{2} + 5\right\}}$$

$$= 2\pi i \frac{6}{\frac{1}{4}+3} = \frac{48\pi i}{13}$$

$$\boxed{B. \frac{48\pi i}{13}}$$

Q5.16.2 MCQ | 2016 | 1M | Ans: B | ● ● ●

$$f(z) = z + z^*$$

Let,

$$z = x + iy$$

$$f(z) = 2x$$

$$f(z) = x + iy + x - iy$$

$$u + iv = 2x$$

$$u = 2x$$

$$v = 0$$

For analytic,

$$u_x = v_y \qquad \& \qquad u_y = -v_x$$

$$u_x = \frac{du}{dx} = 2$$

$$v_y = 0$$

$$v_x = 0$$

$$u_y = 0$$

Since

$$u_x \neq v_y$$

So it is not analytic.

$$f(z) = 2x$$

This function is continuous.

$$\boxed{\text{B. f(z) is continuous but not analytic.}}$$

Q5.15.1 MCQ 2015 1M Ans: B ● ○ ○

Given a complex function $f(z)$ defined as the sum of two other complex functions, $g(z)$ and $h(z)$, such that

$$f(z) = g(z) + h(z).$$

Differentiability Rules for Sums of Complex Functions In complex analysis, a fundamental property states that if two complex functions, say $g(z)$ and $h(z)$, are both differentiable at a point z_0, then their sum,

$$f(z) = g(z) + h(z),$$

is also differentiable at z_0. Furthermore, the derivative of the sum is the sum of the derivatives:

$$f'(z_0) = g'(z_0) + h'(z_0).$$

If $g'(z_0)$ and $h'(z_0)$ exist, then $f'(z_0)$ exists and

$$f'(z_0) = g'(z_0) + h'(z_0).$$

Option A: If $f(z)$ is differentiable at z_0, then $g(z)$ and $h(z)$ are also differentiable at z_0.

False. While $f(z)$ being differentiable implies certain conditions on its components, it does not guarantee that each individual component function must be differentiable. For example, consider

$$f(z) = z.$$

This function is differentiable everywhere. We can write

$$f(z) = \mathrm{Re}(z) + i \cdot \mathrm{Im}(z).$$

Let

$$g(z) = \mathrm{Re}(z) = x$$

and

$$h(z) = i \cdot \mathrm{Im}(z) = iy.$$

Here,

$$f(z) = g(z) + h(z).$$

While $f(z)$ and $h(z)$ are differentiable, $g(z) = x$ is not complex differentiable (its partial derivative with respect to y is zero, violating the Cauchy–Riemann equations). Therefore, knowing $f(z)$ is differentiable doesn't force both $g(z)$ and $h(z)$ to be differentiable.

Option B: If $g(z)$ and $h(z)$ are differentiable at z_0, then $f(z)$ is also differentiable at z_0.

True. Following sum rule for differentiation in complex analysis: If the derivatives $g'(z_0)$ and $h'(z_0)$ exist, then the derivative of their sum, $f'(z_0)$, also exists and is equal to

$$g'(z_0) + h'(z_0).$$

Option C: If $f(z)$ is continuous at z_0, then it is differentiable at z_0.

False.A function must be continuous at a point to be differentiable there, but the reverse is not true. Example,

$$f(z) = |z|,$$

which is continuous everywhere but not differentiable at

$$z = 0.$$

Option D: If $f(z)$ is differentiable at z_0, then so are its real and imaginary parts. If

$$f(z) = u(x, y) + iv(x, y),$$

differentiability of $f(z)$ implies that the real-valued functions $u(x, y)$ and $v(x, y)$ are real differentiable (i.e., their partial derivatives exist and satisfy the Cauchy–Riemann equations).

However, the statement might imply that the complex-valued functions $u(x, y)$ and $v(x, y)$ are themselves complex differentiable. This is not necessarily true. For

$$f(z) = z = x + iy,$$

we have

$$u(x, y) = x, \qquad v(x, y) = y.$$

Neither u nor v is complex differentiable on its own.

$\boxed{\text{B. If } g(z) \text{ and } h(z) \text{ are differentiable at } z_0\text{, then } f(z) \text{ is also differentiable at } z_0.}$

Q5.14.1 MCQ 2014 1M Ans: C ● ○ ○

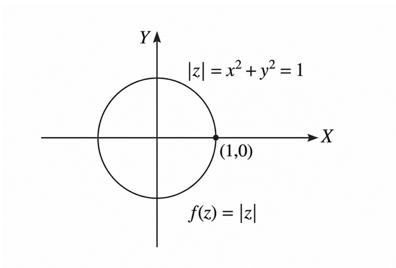

Fig. Solution Q5.14.1

$$z = x + \text{iy}$$
$$z^* = x - \text{iy}$$
$$zz^* = (x + \text{iy})(x - \text{iy})$$
$$= x^2 + y^2$$

Also,

$$|z| = 1$$
$$zz^* = x^2 + y^2$$
$$\boxed{\text{C. The point (1,0)}}$$

Q5.14.2 MCQ 2014 1M Ans: B ● ● ○

Assume the given expression is:

$$x = 1^i$$

$$1 = \cos(2n\pi) + i\sin(2n\pi) = e^{i2n\pi}$$

$$x = \left(e^{i2n\pi}\right)^i$$

$$x = e^{i^2 \cdot 2n\pi}$$

$$x = e^{-2n\pi}$$

Let us analyze the nature of the resulting values for $e^{-2n\pi}$:

- **Real values:** Since the base e is a positive real mathematical constant and the exponent $-2n\pi$ is purely real for any integer n, the value of $e^{-2n\pi}$ contains no imaginary component. Thus, all values are real.
- **Non-negative values:** The exponential function e^y is strictly positive ($e^y > 0$) for any real exponent y. Therefore, $e^{-2n\pi}$ is always strictly positive, making all values non-negative.

Consequently, all values of 1^i are real and non-negative.

$$\boxed{\text{B. real and non -negative}}$$

Key Points

- **Multi-valued Functions:** Powers of complex numbers z^w are generally multi-valued because they depend on the complex logarithm: $z^w = e^{w\log(z)} = e^{w[\ln|z|+i(\theta+2n\pi)]}$.
- **Exponential Function Behavior:** For any real domain input k, the real-valued function $f(k) = e^k$ maps entirely onto the strictly positive open interval $(0, \infty)$.

Mistakes to be avoided

Do not assume $1^i = 1$ is the unique solution; ignoring the general argument $2n\pi$..

Q5.14.3 MCQ 2014 2M Ans: C ● ○ ○

The given function is:

$$f(z) = \frac{z^2}{z^2 - 1} = \frac{z^2}{(z-1)(z+1)}$$

The singularities of $f(z)$ are simple poles:

$$z^2 - 1 = 0 \implies z = 1, -1$$

The given contour is a circle described by:

$$C : |z - 1| = 1$$

This represents a circle centered at $z = 1$ with a radius of 1.

- For $z = 1$:
$$|1 - 1| = 0 < 1 \implies z = 1 \text{ lies inside the curve } C.$$
- For $z = -1$:
$$|-1 - 1| = |-2| = 2 > 1 \implies z = -1 \text{ lies outside the curve } C.$$

By Cauchy's Residue Theorem,

$$\oint_C f(z)\,dz = 2\pi i\,\text{Res}[f(z),1]$$

Residue at the simple pole $z = 1$:

$$\text{Res}[f(z),1] = \lim_{z\to 1}(z-1)\frac{z^2}{(z-1)(z+1)} = \lim_{z\to 1}\frac{z^2}{z+1}$$

$$\text{Res}[f(z),1] = \frac{1^2}{1+1} = \frac{1}{2}$$

$$\oint_C f(z)\,dz = 2\pi i\left[\frac{1}{2}\right] = \pi i$$

$$\boxed{C.\pi i}$$

Q5.13.1 MCQ 2013 1M Ans: B ●●○

Let us assume:

$$x = \sqrt{-i}$$

In polar form, using the principal argument $\theta = -\frac{\pi}{2}$:

$$-i = \cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right) = e^{-i\frac{\pi}{2}}$$

Taking the square root to find the first root (x_1):

$$x = \left(e^{-i\frac{\pi}{2}}\right)^{\frac{1}{2}}$$

$$x = e^{-i\frac{\pi}{4}} = \cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)$$

Using $\theta = \frac{3\pi}{2}$ to find the second root (x_2):

$$-i = \cos\left(\frac{3\pi}{2}\right) + i\sin\left(\frac{3\pi}{2}\right) = e^{i\frac{3\pi}{2}}$$

Taking the square root:

$$x = \left(e^{i\frac{3\pi}{2}}\right)^{\frac{1}{2}} = e^{i\frac{3\pi}{4}}$$

$$x = \cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)$$

$$\boxed{B.\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right),\ \cos\left(\frac{3\pi}{4}\right) + i\sin\left(\frac{3\pi}{4}\right)}$$

Q5.13.2 MCQ 2013 2M Ans: A ●○○

The given contour integral is:

$$\oint_C \frac{z^2-4}{z^2+4}\,dz \quad \text{where} \quad C: |z-i| = 2$$

$$f(z) = \frac{z^2-4}{(z-2i)(z+2i)}$$

The singularities of $f(z)$ are simple poles at:

$$z = 2i \quad \text{and} \quad z = -2i$$

The given contour C is a circle centered at i (or $(0,1)$ in the Cartesian plane) with a radius of 2. Let us test which poles lie inside this contour:

- For $z = 2i$:

$$|2i - i| = |i| = 1 < 2 \implies z = 2i \text{ lies inside the curve } C.$$

- For $z = -2i$:

$$|-2i - i| = |-3i| = 3 > 2 \implies z = -2i \text{ lies outside the curve } C.$$

By Cauchy's Residue Theorem, the value of the integral depends only on the residue of the pole inside the contour:

$$\oint_C \frac{z^2 - 4}{z^2 + 4}\, dz = 2\pi i\, \mathrm{Res}\{f(z), 2i\}$$

Finding the residue at the simple pole $z = 2i$:

$$\mathrm{Res}\{f(z), 2i\} = \lim_{z \to 2i} (z - 2i)\frac{z^2 - 4}{(z - 2i)(z + 2i)} = \frac{(2i)^2 - 4}{2i + 2i}$$

$$\mathrm{Res}\{f(z), 2i\} = \frac{4i^2 - 4}{4i}$$

$$\mathrm{Res}\{f(z), 2i\} = \frac{-4 - 4}{4i} = \frac{-8}{4i} = -\frac{2}{i}$$

Therefore,

$$\oint_C \frac{z^2 - 4}{z^2 + 4}\, dz = 2\pi i\left(-\frac{2}{i}\right) = -4\pi$$

$$\boxed{A. - 4\pi}$$

Q5.12.1 MCQ 2012 1M Ans: A ● ○ ○

Given that:

$$x = \sqrt{-1}$$

$$x = \sqrt{i^2} = i$$

The number i lies on the positive imaginary axis, so its magnitude is 1 and its principal argument is $\frac{\pi}{2}$:

$$i = \cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right) = e^{i\frac{\pi}{2}}$$

$$x^x = i^i = \left(e^{i\frac{\pi}{2}}\right)^i$$

$$x^x = e^{i^2\frac{\pi}{2}}$$

$$x^x = e^{-\frac{\pi}{2}}$$

$$\boxed{A. e^{-\frac{\pi}{2}}}$$

Key Points

- **Euler's Formula:** Any complex number can be expressed in exponential form as $z = re^{i\theta}$, where $r = |z|$ and $\theta = \arg(z)$. For $z = i$, $r = 1$ and $\theta = \frac{\pi}{2}$.
- **:** $z^w = e^{w\ln(z)}$.

Q5.11.1 MCQ 2011 1M Ans: D ● ○ ○

The given algebraic equation is:

$$x^3 + x^2 + x + 1 = 0$$
$$(x^3 + x^2) + (x + 1) = 0$$
$$x^2(x + 1) + 1(x + 1) = 0$$
$$(x + 1)(x^2 + 1) = 0$$
$$x + 1 = 0 \implies x = -1$$
$$x^2 + 1 = 0 \implies x^2 = -1 \implies x = \pm\sqrt{-1}$$
$$x = +j, -j$$

Therefore, the roots of the given algebraic equation are $(-1, +j, -j)$.

$$\boxed{D.(-1, +j, -j)}$$

Q5.11.2 MCQ 2011 1M Ans: D ● ● ○

Let $z = x + iy$.
Given that z lies in the first quadrant of the unit circle:

$$0 < x < 1, \quad 0 < y < 1$$
$$0 < \sqrt{x^2 + y^2} < 1$$

Now, consider the reciprocal function $\frac{1}{z}$:

$$\frac{1}{z} = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2} = \left(\frac{x}{x^2 + y^2}\right) + i\left(\frac{-y}{x^2 + y^2}\right)$$

Since $x > 0$ and $y > 0$:

$$\frac{x}{x^2 + y^2} > 0 \quad \text{and} \quad \frac{-y}{x^2 + y^2} < 0$$

Since the real part is positive and the imaginary part is negative, $\frac{1}{z}$ lies in the IV quadrant.
Next, checking the magnitude of $\frac{1}{z}$:

$$\left|\frac{1}{z}\right| = \left|\frac{1}{x + iy}\right| = \frac{1}{\sqrt{x^2 + y^2}}$$

Given that $0 < \sqrt{x^2 + y^2} < 1$, its reciprocal satisfies:

$$\frac{1}{\sqrt{x^2 + y^2}} > 1$$

So, $\frac{1}{z}$ lies outside the unit circle in the IV quadrant.

$$\boxed{D}$$

Key Points

- **Complex Inversion Mapping:** The mapping $w = \frac{1}{z}$ inverts the magnitude ($|w| = \frac{1}{|z|}$) and negates the argument ($\arg(w) = -\arg(z)$).
- **Quadrant Mapping:** Points in the first quadrant ($0 < \theta < \pi/2$) map to the fourth quadrant ($-\pi/2 < -\theta < 0$) under inversion.

Mistakes to be avoided

Note that if $|z| < 1$, then $|1/z| > 1$;the point does not remain inside the unit circle.

Q5.08.1 MCQ 2008 2M Ans: D ● ● ○

The given function is:

$$X(z) = \frac{z}{(z-a)^2} \quad \text{with} \quad |z| > a$$

We need to find the residue of $F(z) = X(z)z^{n-1}$ at $z = a$ for $n \geq 0$. Substituting $X(z)$ into the expression:

$$F(z) = \frac{z}{(z-a)^2} \cdot z^{n-1} = \frac{z^n}{(z-a)^2}$$

The singularity at $z = a$ is a pole of order $m = 2$.

$$\text{Res}\,[F(z), z_0] = \frac{1}{(m-1)!} \lim_{z \to z_0} \frac{d^{m-1}}{dz^{m-1}} [(z-z_0)^m F(z)]$$

Substituting $z_0 = a$ and $m = 2$:

$$\text{Res}\,[F(z), a] = \frac{1}{(2-1)!} \lim_{z \to a} \frac{d}{dz} \left[(z-a)^2 \cdot \frac{z^n}{(z-a)^2}\right]$$

$$\text{Res}\,[F(z), a] = \lim_{z \to a} \frac{d}{dz} [z^n]$$

Computing the derivative with respect to z:

$$\frac{d}{dz} [z^n] = nz^{n-1}$$

Evaluating the limit by substituting $z = a$:

$$\text{Res}\,[F(z), a] = na^{n-1}$$

$$\boxed{D. na^{n-1}}$$

Q5.07.1 MCQ 2007 2M Ans: B ● ● ○

Given integral is:

$$I = \oint_C \frac{dz}{1+z^2} = \oint_C \frac{dz}{(z+i)(z-i)}$$

The integrand $f(z) = \frac{1}{1+z^2}$ has simple poles at:

$$z = -i$$

$$z = i$$

The given contour C is a circle described by:

$$\left|z - \frac{i}{2}\right| = 1$$

Checking the position of the poles with respect to the contour C:

- For $z = i$: $\left|i - \frac{i}{2}\right| = \left|\frac{i}{2}\right| = \frac{1}{2} < 1$ (Lies inside C)
- For $z = -i$: $\left|-i - \frac{i}{2}\right| = \left|-\frac{3i}{2}\right| = \frac{3}{2} > 1$ (Lies outside C)

Therefore, $z = i$ will lie inside contour C.

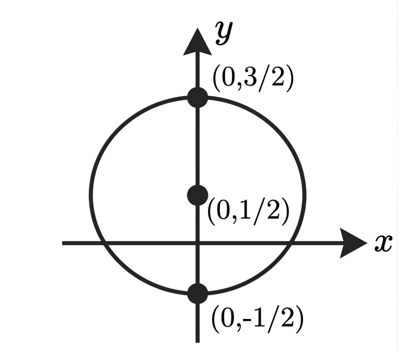

Fig. Solution Q5.07.1

By Cauchy's Residue Theorem:

$$\oint_C \frac{dz}{1+z^2} = 2\pi i\,[\text{Res } f(z) \text{ at } z = i]$$

$$\oint_C \frac{dz}{1+z^2} = 2\pi i\left\{\frac{1}{i+i}\right\} = 2\pi i\left\{\frac{1}{2i}\right\} = \pi$$

$$\boxed{B.\pi}$$

Q5.97.1 MCQ 1997 1M Ans: B ● ○ ○

The complex exponential function e^z is periodic with a fundamental period of 2πi

Because $e^{z+2\pi \mathrm{i}} = e^z$, the function's values repeat identically every time you add an integer multiple of 2πi (i.e., $2\pi n$i where n is any integer).

$$\boxed{B.\ 2\pi i}$$

Q5.26.1 MCQ 2026 1M Ans: A ● ○ ○

Given the function:

$$f(z) = z^2 + z + 1$$

We need to evaluate the contour integral:

$$I = \oint_\gamma \frac{f(z)}{z-1}\,dz = \oint_\gamma \frac{z^2+z+1}{z-1}\,dz$$

where γ is a simple closed contour enclosing the point $z = 1 + 0j = 1$. The integrand has a simple pole at $z = 1$. Since the contour γ explicitly encloses the point $z = 1$, this pole lies inside the region bounded by γ.

According to Cauchy's Integral Formula, if a function $f(z)$ is analytic within and on a simple closed contour γ, and a is a point inside γ, then:

$$\oint_\gamma \frac{f(z)}{z-a}\,dz = 2\pi j \cdot f(a)$$

Here, $a = 1$. Evaluating $f(z)$ at $z = 1$:

$$f(1) = 1^2 + 1 + 1 = 3$$

Substituting $f(1)$ back into the formula:

$$I = 2\pi j \cdot f(1) = 2\pi j \cdot 3 = 6\pi j$$

Thus, the value of the integral is $6\pi j$.

$$\boxed{A.\ 6\pi j}$$

Key Points

- **Cauchy's Integral Formula:** $\oint_\gamma \frac{f(z)}{z-a}\,dz = 2\pi j \cdot f(a)$ when the point a lies inside the contour γ.
- If the singularity lay outside the contour, the integral would be 0 by Cauchy's Integral Theorem.

Q5.25.1 NAT 2025 1M Ans: 2 ● ○ ○

Given the function:

$$f(z) = \frac{2z+1}{z^2 - z}$$

To find the singular points (poles), we equate the denominator to zero:

$$z^2 - z = 0 \implies z(z-1) = 0$$

Thus, the singular points are simple poles at $z = 0$ and $z = 1$.
The residue at a simple pole $z = a$ is given by:

$$\text{Res}(f, a) = \lim_{z\to a}(z-a)f(z)$$

Evaluating the residue at $z = 0$:

$$\text{Res}(f,0) = \lim_{z\to 0} z \cdot \frac{2z+1}{z(z-1)} = \lim_{z\to 0}\frac{2z+1}{z-1} = \frac{2(0)+1}{0-1} = -1$$

Evaluating the residue at $z = 1$:

$$\text{Res}(f,1) = \lim_{z\to 1}(z-1) \cdot \frac{2z+1}{z(z-1)} = \lim_{z\to 1}\frac{2z+1}{z} = \frac{2(1)+1}{1} = 3$$

The sum of the residues at all singular points is:

$$\text{Sum of residues} = \text{Res}(f,0) + \text{Res}(f,1) = -1 + 3 = 2$$

$$\boxed{2}$$

Key Points

Here, $\lim_{z\to\infty} z \cdot \frac{2z+1}{z^2-z} = \lim_{z\to\infty} \frac{2z^2+z}{z^2-z} = 2$, which verifies our calculation.

Q5.25.2 MSQ 2025 2M Ans: A,B ● ● ●

Analyse each statement based on Cauchy's Integral Theorem:

Statement A: Cauchy's theorem cannot be directly applied to conclude that $\oint_C f(z)\,dz = 0$ when $f(z) = \frac{1}{z^2}$ and C is the unit circle. Cauchy's Integral Theorem requires the function to be analytic everywhere inside and on the closed contour C. For $f(z) = \frac{1}{z^2}$, the function has a singularity at $z = 0$, which lies inside the unit circle. Thus, the theorem cannot be directly applied. Statement A is correct.

Statement B: If $f(z)$ is analytic in D, then it can be concluded that $\oint_C f(z)\,dz = 0$ for any simple closed path C in D. This is the direct definition of Cauchy's Integral Theorem for a simply connected domain. Since $f(z)$ is analytic everywhere within D, any closed contour C within D will yield an integral value of zero. Statement B is correct.

Statement C: The function $f(z)$ must be analytic in D to conclude $\oint_C f(z)\,dz = 0$ for any simple closed path C in D. The converse of Cauchy's Theorem states, if $f(z)$ is continuous in a domain D and $\oint_C f(z)\,dz = 0$ for every simple closed contour C in D, then $f(z)$ must be analytic in D. Therefore, analyticity is a necessary and sufficient condition for the integral to vanish along every arbitrary closed path. Statement C is correct,but it's more of a condition than an independent statement that needs separate consideration beyond Cauchy's theorem.

Statement D: $\oint_C f(z)\,dz \neq 0$ when $f(z) = \frac{1}{z^2}$, since the function is not analytic at $z = 0$. Even though the function is not analytic at $z = 0$, the value of this specific integral over the unit circle is found via the residue theorem. The residue of $\frac{1}{z^2}$ at $z = 0$ is 0 (since the coefficient of the $\frac{1}{z}$ term in its Laurent series expansion is zero). Therefore, $\oint_C \frac{1}{z^2}\,dz = 2\pi i(0) = 0$. Thus, the assertion that the integral is non-zero is false. Statement D is incorrect.

A . Cauchy's theorem cannot be directly applied to conclude that $\oint_C f(z)\,dz = 0$ when $f(z) = \frac{1}{z^2}$ and C is the unit circle.

B.If $f(z)$ is analytic in D, then it can be concluded that $\oint_C f(z)\,dz = 0$ for any simple closed path C in D.

Key Points

- **Cauchy-Goursat Theorem:** If $f(z)$ is analytic at all points interior to and on a simple closed contour C, then $\oint_C f(z)\,dz = 0$.
- **Morera's Theorem:** If $f(z)$ is continuous in a domain D and $\oint_C f(z)\,dz = 0$ for any closed contour C, then $f(z)$ is analytic in D.

Q5.24.1 MCQ 2024 1M Ans: B ● ● ○

Given the complex variable $z = x + iy$, its complex conjugate is $\bar{z} = x - iy$. We are given the algebraic equation:

$$\bar{z}^2 + z^2 = 2$$

Let us substitute the expressions for z and $\bar{z}$ into the equation:

$$(x - iy)^2 + (x + iy)^2 = 2$$

$$(x^2 - 2ixy + i^2y^2) + (x^2 + 2ixy + i^2y^2) = 2$$

$$(x^2 - 2ixy - y^2) + (x^2 + 2ixy - y^2) = 2$$

$$2x^2 - 2y^2 = 2$$

$$x^2 - y^2 = 1$$

This equation matches the standard conic section equation for a rectangular hyperbola, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ where $a = b = 1$.

$$\boxed{B.\ Hyperbola}$$

Key Points

Conic Section Identifications:
$x^2 + y^2 = r^2$ represents a circle. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ represents an ellipse. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ represents a hyperbola.

Q5.24.2 MSQ 2024 2M Ans: A,C ● ● ●

Given that $f(z) = u(x, y) + iv(x, y)$ is analytic, it must satisfy the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

We are also given that the complex conjugate function $\overline{f(z)} = u(x, y) - iv(x, y)$ is analytic. Let us denote $\overline{f(z)} = U(x, y) + iV(x, y)$, where:

$$U(x, y) = u(x, y) \quad \text{and} \quad V(x, y) = -v(x, y)$$

Since $\overline{f(z)}$ is analytic, it must also satisfy the Cauchy-Riemann equations:

$$\frac{\partial U}{\partial x} = \frac{\partial V}{\partial y} \implies \frac{\partial u}{\partial x} = -\frac{\partial v}{\partial y}$$

$$\frac{\partial U}{\partial y} = -\frac{\partial V}{\partial x} \implies \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$$

Comparing the two sets of conditions simultaneously: From $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial x} = -\frac{\partial v}{\partial y}$:

$$\frac{\partial u}{\partial x} = 0 \quad \text{and} \quad \frac{\partial v}{\partial y} = 0$$

From $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ and $\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$:

$$\frac{\partial u}{\partial y} = 0 \quad \text{and} \quad \frac{\partial v}{\partial x} = 0$$

Thus, all first-order partial derivatives are zero, which validates Statement A:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 0$$

The derivative of an analytic function can be expressed as:

$$\frac{df(z)}{dz} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$$

Substituting the values $\frac{\partial u}{\partial x} = 0$ and $\frac{\partial v}{\partial x} = 0$:

$$\frac{df(z)}{dz} = 0 + i(0) = 0$$

This validates Statement C.

$$\boxed{A.\ \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 0, C.\ \frac{df(z)}{dz} = 0}$$

Q5.23.1 MCQ 2023 2M Ans: A ● ● ○

Given the function:

$$f(z) = j\frac{1-z}{1+z}$$

Let $w = f(z)$. To find the mapping properties of the inverse function $f^{-1}(z)$, which maps the real axis of the w-plane back into the z-plane. Let w be a real number. Therefore, $w = \bar{w}$.

$$j\frac{1-z}{1+z} = \overline{\left(j\frac{1-z}{1+z}\right)}$$

Since $\bar{j} = -j$, evaluating the right-hand side yields:

$$j\frac{1-z}{1+z} = -j\frac{1-\bar{z}}{1+\bar{z}}$$

$$\frac{1-z}{1+z} = -\frac{1-\bar{z}}{1+\bar{z}}$$

$$(1-z)(1+\bar{z}) = -(1-\bar{z})(1+z)$$

$$1+\bar{z}-z-z\bar{z} = -(1+z-\bar{z}-z\bar{z})$$

$$1+\bar{z}-z-z\bar{z} = -1-z+\bar{z}+z\bar{z}$$

$$1-z\bar{z} = -1+z\bar{z}$$

$$2 = 2z\bar{z} \implies z\bar{z} = 1$$

Since $z\bar{z} = |z|^2$:

$$|z|^2 = 1 \implies |z| = 1$$

The equation $|z| = 1$ represents a unit circle centered at the origin in the complex z-plane.

$$\boxed{A.\text{unit circle with centre at the origin.}}$$

Q5.22.1 MSQ 2022 2M Ans: A,C ● ● ○

Given the complex number:

$$z = \frac{a+jb}{a-jb}$$

where $a > 0$ and $b > 0$.

Part 1: Magnitude Calculation The magnitude of a quotient of two complex numbers is the quotient of their magnitudes:

$$|z| = \left|\frac{a+jb}{a-jb}\right| = \frac{|a+jb|}{|a-jb|}$$

Using the formula for the magnitude of a complex number $|x+jy| = \sqrt{x^2+y^2}$:

$$|a+jb| = \sqrt{a^2+b^2}$$

$$|a-jb| = \sqrt{a^2+(-b)^2} = \sqrt{a^2+b^2}$$

Therefore:

$$|z| = \frac{\sqrt{a^2+b^2}}{\sqrt{a^2+b^2}} = 1$$

Statement C is correct and Statement D is incorrect.

Part 2: Phase Calculation The phase angle (θ) of a quotient of two complex numbers is the difference between the phase of the numerator and the phase of the denominator:

$$\angle z = \angle(a+jb) - \angle(a-jb)$$

Since $a > 0$ and $b > 0$, the numerator $a + jb$ lies in the first quadrant:

$$\angle(a + jb) = \tan^{-1}\left(\frac{b}{a}\right)$$

The denominator $a - jb$ lies in the fourth quadrant:

$$\angle(a - jb) = \tan^{-1}\left(\frac{-b}{a}\right) = -\tan^{-1}\left(\frac{b}{a}\right)$$

Substituting these phase angles back into the relation:

$$\angle z = \tan^{-1}\left(\frac{b}{a}\right) - \left[-\tan^{-1}\left(\frac{b}{a}\right)\right] = 2\tan^{-1}\left(\frac{b}{a}\right)$$

Statement A is correct and Statement B is incorrect.

$$\boxed{\text{A. The phase is } 2\tan^{-1}\left(\frac{b}{a}\right), \text{ C. The magnitude is 1}}$$

Key Points

- For any complex number w, $|w| = |\bar{w}|$.
- The phase of a ratio of complex numbers satisfies $\angle\left(\frac{w_1}{w_2}\right) = \angle w_1 - \angle w_2$.

Q5.22.2 | NAT | 2022 | 2M | Ans: 0.49 to 0.51 | ● ● ○

Given the function:

$$f(z) = \frac{1}{(z+1)(z+2)(z+3)}$$

To find the residue of $f(z)$ at the simple pole $z = -1$. The residue at a simple pole $z = a$ is given by:

$$\text{Res}(f, a) = \lim_{z\to a}(z - a)f(z)$$

Setting $a = -1$:

$$\text{Res}(f, -1) = \lim_{z\to -1}(z+1)\cdot\frac{1}{(z+1)(z+2)(z+3)}$$

$$\text{Res}(f, -1) = \lim_{z\to -1}\frac{1}{(z+2)(z+3)}$$

$$\text{Res}(f, -1) = \frac{1}{(-1+2)(-1+3)} = \frac{1}{(1)(2)} = \frac{1}{2} = 0.5$$

$$\boxed{0.5}$$

Q5.21.1 | NAT | 2021 | 1M | Ans: 0 | ● ● ○

Given the function:

$$f(z) = \frac{1}{z^2 + 6z + 9}$$

To find the singular points by setting the denominator to zero:

$$z^2 + 6z + 9 = 0 \implies (z+3)^2 = 0 \implies z = -3$$

Thus, the function has a pole of order 2 at $z = -3$. The integration path is a circle c centered at the origin with a unit radius:

$$|z| = 1$$

The distance of the pole from the origin is $|-3| = 3$. Since $3 > 1$, the singularity lies entirely outside the unit circle contour c.

According to Cauchy's Integral Theorem, if a function $f(z)$ is analytic at all points interior to and on a simple closed contour c, then the contour integral along that path is zero:

$$\oint_c f(z)\,dz = 0$$

$$\boxed{0}$$

Q5.21.2 MCQ | 2021 | 2M | Ans: B | ● ● ●

Given series expansion is:

$$f(z) = (z-1)^{-1} - 1 + (z-1) - (z-1)^2 + \cdots$$

Rewriting the series:

$$f(z) = \frac{1}{z-1} - \left[1 - (z-1) + (z-1)^2 - \cdots\right]$$

The expression inside the brackets is an infinite geometric series of the form:

$$1 - x + x^2 - x^3 + \cdots = \frac{1}{1+x}$$

where $x = z - 1$. This series converges when $|x| < 1$, which means $|z-1| < 1$. Substituting $x = z - 1$ back into the geometric series sum formula,

$$1 - (z-1) + (z-1)^2 - \cdots = \frac{1}{1+(z-1)} = \frac{1}{z}$$

Substitute this closed-form sum back into the expression for $f(z)$,

$$f(z) = \frac{1}{z-1} - \frac{1}{z}$$

Hence,

$$f(z) = \frac{z-(z-1)}{z(z-1)} = \frac{1}{z(z-1)}$$

$$\boxed{B.\frac{1}{z(z-1)} \text{ for } |z-1| < 1}$$

Key Points

- **Geometric Series Convergence:** The expansion $\frac{1}{1+x} = \sum_{n=0}^{\infty}(-1)^n x^n$ is valid if and only if $|x| < 1$.
- This representation is the Laurent series expansion of the function $f(z) = \frac{1}{z(z-1)}$ valid in the punctured disk region $0 < |z-1| < 1$.

Q5.20.1 MCQ | 2020 | 1M | Ans: B | ● ● ○

Given the function:

$$f(z) = \frac{1}{z+a} \quad \text{where } a > 0$$

To find the value of the contour integral $\oint_C f(z)\,dz$ over a circle C centered at $(-a, 0)$ with a radius $R > 0$ in the anti-clockwise direction. Pole,

$$z + a = 0 \implies z = -a$$

The center of the circle C corresponds to the complex number $z = -a$. Since the radius is $R > 0$, the simple pole at $z = -a$ always lies inside the contour circle C no matter what R is. By Cauchy's Integral Formula:

$$\oint_C \frac{1}{z - z_0}\, dz = 2\pi i$$

where z_0 lies inside the contour C. Here, $z_0 = -a$. Since the contour is traversed in the positive (anti-clockwise) direction, the integral evaluates to:

$$\oint_C \frac{1}{z + a}\, dz = 2\pi i$$

$$\boxed{B.\ 2\pi i}$$

Q5.19.1 MCQ 2019 2M Ans: C ● ● ●

Given that a complex function $f(z) = u(x, y) + iv(x, y)$ and its complex conjugate $f^*(z) = u(x, y) - iv(x, y)$ are both analytic in the entire complex plane. For $f(z)$ to be analytic, it must satisfy the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

For $f^*(z) = u + i(-v)$ to be analytic, its component functions must also satisfy the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial(-v)}{\partial y} \implies \frac{\partial u}{\partial x} = -\frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial(-v)}{\partial x} \implies \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$$

Comparing the two sets of equations simultaneously: From $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial x} = -\frac{\partial v}{\partial y}$:

$$\frac{\partial u}{\partial x} = 0 \quad \text{and} \quad \frac{\partial v}{\partial y} = 0$$

From $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ and $\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$:

$$\frac{\partial u}{\partial y} = 0 \quad \text{and} \quad \frac{\partial v}{\partial x} = 0$$

Since all first-order partial derivatives of u and v are zero everywhere, both $u(x, y)$ and $v(x, y)$ must be real constants. Hence, the function $f(z)$ reduces to a constant function.

$$\boxed{C.\ f(z) = \text{constant}}$$

Key Points

- If a function and its complex conjugate are both analytic over a region, the function must be a constant throughout that region.

Mistakes to be avoided

- Avoid choosing basic algebraic structures like $x^2 + y^2$, as they are real-valued but not analytic because they fail the Cauchy-Riemann conditions.

Q5.18.1 | MCQ | 2018 | 1M | Ans: B | ● ● ○

Given two functions:

$$f_1(z) = z^2$$

$$f_2(z) = \bar{z}$$

Analyticity of $f_1(z) = z^2$: Any polynomial function of z (such as $z, z^2, z^3, \ldots$) is analytic everywhere in the complex plane (entire function).Verify via partial derivatives:

$$f_1(z) = (x + iy)^2 = (x^2 - y^2) + i(2xy)$$

Here, $u = x^2 - y^2$ and $v = 2xy$.

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial y} = 2x \implies \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -2y, \quad \frac{\partial v}{\partial x} = 2y \implies \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

The Cauchy-Riemann equations are satisfied everywhere, so $f_1(z)$ is analytic. Analyticity of $f_2(z) = \bar{z}$: Let $f_2(z) = x - iy$. Here, $u = x$ and $v = -y$. Evaluating the partial derivatives:

$$\frac{\partial u}{\partial x} = 1, \quad \frac{\partial v}{\partial y} = -1$$

Since $1 \neq -1$, $\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$. The Cauchy-Riemann equations are violated everywhere. Thus, $f_2(z)$ is not analytic anywhere. Therefore, only $f_1(z)$ is analytic.

$$\boxed{B.\ Only\ f_1(z)\ is\ analytic.}$$

Q5.17.1 | MCQ | 2017 | 1M | Ans: B | ● ● ○

To evaluate the complex conjugate of the cosine function: $\overline{\cos z}$. The complex definition of the cosine function is given by:

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

Taking the complex conjugate of both sides:

$$\overline{\cos z} = \overline{\left(\frac{e^{iz} + e^{-iz}}{2}\right)} = \frac{\overline{e^{iz}} + \overline{e^{-iz}}}{2}$$

Since, $\overline{e^w} = e^{\bar{w}}$,

$$\overline{e^{iz}} = e^{\overline{iz}} = e^{-i\bar{z}}$$

$$\overline{e^{-iz}} = e^{\overline{-iz}} = e^{i\bar{z}}$$

Substituting,

$$\overline{\cos z} = \frac{e^{-i\bar{z}} + e^{i\bar{z}}}{2} = \frac{e^{i\bar{z}} + e^{-i\bar{z}}}{2}$$

Which matches the standard definition of the cosine function evaluated at $\bar{z}$:

$$\cos \bar{z} = \frac{e^{i\bar{z}} + e^{-i\bar{z}}}{2}$$

Thus, $\overline{\cos z} = \cos \bar{z}$.

$$\boxed{B. \cos \bar{z}}$$

Key Points

- **Conjugate Reflection Property:** For any analytic function $f(z)$ that takes real values on the real axis, $\overline{f(z)} = f(\bar{z})$.

Mistakes to be avoided

- Do not confuse the conjugate operator with changing the internal functional sign improperly, such as expecting $\cos z$ or $\sin z$ without checking the expansion definitions.

Q5.16.1 NAT 2016 2M Ans: 1 ●●○

To evaluate the following,

$$I = \frac{1}{2\pi j} \oint_C \frac{z^2+1}{z^2-1}\, dz$$

To find the poles,

$$z^2 - 1 = (z-1)(z+1)$$

The simple poles are at $z = 1$ and $z = -1$. The contour C is a unit circle centered at $1 + 0j = 1$. Its equation is:

$$|z-1| = 1$$

- For $z = 1$: $|1-1| = 0 < 1 \implies z = 1$ lies inside the contour C.
- For $z = -1$: $|-1-1| = |-2| = 2 > 1 \implies z = -1$ lies outside the contour C.

Apply Cauchy's Integral Formula to $z = 1$:

$$\oint_C \frac{\frac{z^2+1}{z+1}}{z-1}\, dz$$

Let $f(z) = \frac{z^2+1}{z+1}$, which is analytic within and on the contour C. By Cauchy's Integral Formula:

$$\oint_C \frac{f(z)}{z-1}\, dz = 2\pi j \cdot f(1)$$

Finding $f(1)$:

$$f(1) = \frac{1^2+1}{1+1} = \frac{2}{2} = 1$$

Substituting this back into our calculation:

$$\oint_C \frac{z^2+1}{z^2-1}\, dz = 2\pi j \cdot (1) = 2\pi j$$

Substitute the contour integral value back into the original expression:

$$I = \frac{1}{2\pi j} \cdot (2\pi j) = 1$$

$\boxed{1}$

Q5.15.1 MCQ 2015 1M Ans: B ● ○ ○

Given the integral:

$$I = \oint_C \frac{1}{z^2}\, dz$$

where the contour C is the unit circle $|z| = 1$ traversed in the clockwise direction. Pole of order 2 at $z = 0$, which lies inside the unit circle. The residue at a pole of order n at $z = a$ is given by:

$$\text{Res}(f, a) = \frac{1}{(n-1)!} \lim_{z \to a} \frac{d^{n-1}}{dz^{n-1}} \left[(z-a)^n f(z)\right]$$

For $n = 2$ and $a = 0$:

$$\text{Res}(f, 0) = \frac{1}{1!} \lim_{z \to 0} \frac{d}{dz} \left[z^2 \cdot \frac{1}{z^2}\right] = \lim_{z \to 0} \frac{d}{dz}(1) = 0$$

Since the residue at the only singularity inside the contour is zero, the value of the integral is zero regardless of clockwise or counter-clockwise.

$$\boxed{B.0}$$

Key Points

- **Residue of z^{-n}:** For any integer $n \neq 1$, the residue of $f(z) = z^{-n}$ at $z = 0$ is exactly zero because there is no $\frac{1}{z}$ term in its Laurent series.

Q5.13.1 MCQ 2013 1M Ans: D ● ● ● VIDEO SOLUTION

The complex function is given by:

$$\tanh(s) = \frac{\sinh(s)}{\cosh(s)}$$

A complex function is analytic everywhere except at points where its denominator is zero (singularities). Therefore, $\tanh(s)$ is not analytic where:

$$\cosh(s) = 0$$

Substituting $s = j\omega$, where ω is a real number,

$$\cosh(j\omega) = \cos(\omega) = 0$$

The cosine function vanishes at odd multiples of $\frac{\pi}{2}$:

$$\omega = (2n+1)\frac{\pi}{2} \quad \text{for } n \in \mathbb{Z}$$

Since $\omega = \text{Im}(s)$, the function is analytic everywhere in the region as long as the imaginary part avoids these singular points:

$$\text{Im}(s) \neq \frac{(2n+1)\pi}{2}$$

$$\boxed{D.\text{Im}(s) \neq \frac{(2n+1)\pi}{2}}$$

Key Points

Hyperbolic to Trigonometric Identities: $\cosh(j\omega) = \cos(\omega)$ and $\sinh(j\omega) = j\sin(\omega)$.

Q5.11.1 MCQ 2011 1M Ans: C ● ○ ○

To evaluate the contour integral:

$$I = \oint_C e^{1/z}\, dz$$

where C is the counter-clockwise unit circle $|z| = 1$.
Essential singularity at $z = 0$, which lies inside the unit circle.
To find the residue at an essential singularity, we calculate the Laurent series expansion of the function around $z = 0$:

$$e^w = 1 + w + \frac{w^2}{2!} + \frac{w^3}{3!} + \cdots$$

Substituting $w = \frac{1}{z}$:

$$e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + \cdots$$

The residue at $z = 0$ is the coefficient of the $\frac{1}{z}$ term in this expansion:

$$\text{Res}(f, 0) = 1$$

Applying Cauchy's Residue Theorem for a counter-clockwise path:

$$I = 2\pi\sqrt{-1} \times \text{Res}(f, 0) = 2\pi\sqrt{-1} \cdot (1) = 2\pi\sqrt{-1}$$

$$\boxed{C.2\pi j}$$

Q5.10.1 MCQ 2010 1M Ans: D ● ● ○

The contour C is $x^2 + y^2 = 16$ is circle centered at the origin with a radius of 4. The contour is traversed in the anti-clockwise direction. To evaluate the integral,

$$I = \oint_C \frac{z^2 + 8}{0.5z - 1.5j}\, dz$$

$$I = \oint_C \frac{z^2 + 8}{0.5(z - 3j)}\, dz = 2\oint_C \frac{z^2 + 8}{z - 3j}\, dz$$

One simple pole at $z = 3j$. The distance of this pole from the origin is $|3j| = 3$. Since $3 < 4$, the pole lies completely inside the contour C. If, $g(z) = \frac{z^2+8}{z-3j}$ then the residue at $z = 3j$ is:

$$\text{Res}(g, 3j) = (3j)^2 + 8 = -9 + 8 = -1$$

$$I = 2 \times [2\pi j \times \text{Res}(g, 3j)] = 2 \times [2\pi j \times (-1)] = -4\pi j$$

$$\boxed{D. - 4\pi j}$$

Q5.09.1 MCQ 2009 1M Ans: D ● ○ ○

Given the complex variable $z = x + jy$, to find the magnitude of e^{jz}. Substitute z into the exponent:

$$jz = j(x + jy) = jx + j^2y$$

Since $j^2 = -1$:

$$jz = -y + jx$$

Hence,

$$e^{jz} = e^{-y+jx} = e^{-y} \cdot e^{jx}$$

Since,

$$|e^{jz}| = |e^{-y}| \cdot |e^{jx}|$$

Since x and y are real numbers, e^{-y} is a positive real quantity, so $|e^{-y}| = e^{-y}$. By Euler's formula, $|e^{jx}| = |\cos(x) + j\sin(x)| = \sqrt{\cos^2(x) + \sin^2(x)} = 1$. Therefore:

$$|e^{jz}| = e^{-y} \cdot 1 = e^{-y}$$

$$\boxed{D.\, e^{-y}}$$

Key Points

- **Magnitude of Complex Exponentials:** For any real value θ, $|e^{j\theta}| = 1$.
- The magnitude of e^z depends entirely on its real part: $|e^{u+jv}| = e^u$.

Q5.09.2 MCQ 2009 1M Ans: A ● ● ○

To evaluate the contour integral:

$$I = \oint \frac{\sin z}{z}\, dz$$

around a simple closed curve enclosing the origin. Analyzing the singularity at $z = 0$ with the Taylor series expansion of $\sin z$:

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots$$

Dividing by z:

$$\frac{\sin z}{z} = 1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots$$

Because this expansion contains only non-negative powers of z, the point $z = 0$ is a removable singularity. The function $f(z) = \frac{\sin z}{z}$ can be redefined at $z = 0$ to be 1, making it analytic everywhere inside and on the closed curve. By Cauchy's Integral Theorem, if a function is analytic everywhere inside and on a simple closed path, its loop integral is zero:

$$I = 0$$

$$\boxed{A.0}$$

Key Points

- **Removable Singularity:** If $\lim_{z \to a} f(z)$ exists and is finite, then $z = a$ is a removable singularity, and the residue at that point is 0.

Q5.09.3 MCQ 2009 2M Ans: B ● ● ○

Given the algebraic equation:

$$x^3 = j$$

where $j = \sqrt{-1}$. We need to find one of its roots. We know,

$$j = 0 + 1j = \cos\left(\frac{\pi}{2}\right) + j\sin\left(\frac{\pi}{2}\right) = e^{j\pi/2}$$

To find all unique cube roots,by the periodicity of $2k\pi$:

$$x^3 = e^{j\left(\frac{\pi}{2} + 2k\pi\right)} \quad \text{for } k = 0, 1, 2$$

Taking the cube root of both sides:

$$x = e^{j\left(\frac{\pi}{6} + \frac{2k\pi}{3}\right)}$$

Finding the root corresponding to $k = 0$:

$$x_0 = e^{j\pi/6} = \cos\left(\frac{\pi}{6}\right) + j\sin\left(\frac{\pi}{6}\right)$$

$$\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} \quad \text{and} \quad \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

We get,

$$x_0 = \frac{\sqrt{3}}{2} + j\frac{1}{2}$$

$$\boxed{B. \frac{\sqrt{3}}{2} + j\frac{1}{2}}$$

Q5.08.1 MCQ 2008 2M Ans: B ● ● ●

$z = x + j0.1$, where the real part x varies from $-\infty$ to $+\infty$, and the imaginary part is constant at $y = 0.1$. Geometrically, this is a horizontal line in the complex z-plane. To find the locus of $w = \frac{1}{z}$ in the complex plane:

$$w = \frac{1}{x + j0.1} = \frac{x - j0.1}{(x + j0.1)(x - j0.1)} = \frac{x - j0.1}{x^2 + (0.1)^2}$$

Let $w = u + jv$, where u is the real part and v is the imaginary part:

$$u = \frac{x}{x^2 + 0.01}$$

$$v = \frac{-0.1}{x^2 + 0.01}$$

$$x^2 + 0.01 = -\frac{0.1}{v}$$

Substitute this back into the expression for u (we have to eliminate x),

$$u = x\left(-\frac{v}{0.1}\right) = -10xv \implies x = -\frac{u}{10v}$$

Substitute x back into the relation $x^2 + 0.01 = -\frac{0.1}{v}$:

$$\left(-\frac{u}{10v}\right)^2 + 0.01 = -\frac{0.1}{v}$$

$$\frac{u^2}{100v^2} + \frac{1}{100} = -\frac{1}{10v}$$

$$u^2 + v^2 = -10v$$

$$u^2 + v^2 + 10v = 0$$

Completing the square for v:

$$u^2 + (v + 5)^2 = 5^2$$

This equation represents a circle in the w-plane with center: $(u, v) = (0, -5)$ and radius: $R = 5$.

$$\boxed{\text{B}}$$

Q5.07.1 | MCQ | 2007 | 2M | Ans: B | ● ○ ○

As $\lim_{z\to 0}\frac{\sin z}{z}=1$. Therefore the function has $z=0$ is a pole of order 2.

$$\boxed{\text{B. A pole of order 2}}$$

Key Points

A function $f(z)$ has a pole of order n at $z=a$ if $\lim_{z\to a}(z-a)^n f(z)$ is a non-zero finite value.

Q5.05.1 | MCQ | 2005 | 2M | Ans: A | ● ○ ○

$$|z-5-5i|=2$$

$\Rightarrow |z-(5+5i)|=2$ represents a circle of radius 2. and center $(5,5)$

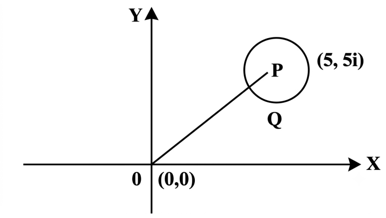

Fig. Solution Q5.05.1

From figure,

$$\text{OP}=\sqrt{5^2+5^2}=5\sqrt{2}$$

OQ is minimum distance from the origin. $\text{OQ}=5\sqrt{2}-2$ as $\text{PQ}=\text{radius}=2$.

$$\boxed{A.\ 5\sqrt{2}-2}$$

Q5.05.2 | MCQ | 2005 | 2M | Ans: C | ● ● ○

Given, $z^3=\bar{z}$ and we know, $z\bar{z}=|z|^2$,

$$\Rightarrow \quad z^4=|z|^2 \text{ (on multiplying z both side)}$$

Now by hit and trial method we see the solution being

$$Z^4=1$$

$$\boxed{C. Z^4=1}$$

Q5.02.1 | MCQ | 2002 | 1M | Ans: A | ● ● ○

Given the bilinear transformation:

$$w=\frac{z-1}{z+1}$$

We want to find where the inside of the unit circle in the z-plane ($|z|<1$) maps to in the w-plane.

$$w(z+1)=z-1 \implies wz+w=z-1$$

$$wz-z=-1-w \implies z(1-w)=1+w$$

$$z=\frac{1+w}{1-w}$$

The boundary of the region is the unit circle, $|z| = 1$:

$$\left|\frac{1+w}{1-w}\right| = 1 \implies |1+w| = |1-w|$$

Let $w = u + iv$:

$$|1+u+iv| = |1-u-iv|$$

$$(1+u)^2 + v^2 = (1-u)^2 + v^2$$

$$1 + 2u + u^2 = 1 - 2u + u^2 \implies 4u = 0 \implies u = 0$$

Thus, the boundary $|z| = 1$ maps to the imaginary axis ($u = 0$) of the w-plane. To check where the interior region $|z| < 1$ maps, let's test a point inside the unit circle, such as the origin $z = 0$:

$$w = \frac{0-1}{0+1} = -1$$

The point $w = -1$ has a real part $u = -1 < 0$, which lies in the left half of the w-plane. Therefore, the inside of the unit circle in the z-plane maps to the left half of the w-plane.

$$\boxed{\text{A. maps the inside of the unit circle in the } z\text{-plane to the left half of the } w\text{-plane.}}$$

Q5.97.1 MCQ 1997 1M Ans: B ● ○ ○

Given the equation:

$$|z+1| = 1$$

If $z = x + jy$

$$|x + jy + 1| = 1$$

$$|(x+1) + jy| = 1$$

Taking the magnitude on both sides:

$$\sqrt{(x+1)^2 + y^2} = 1$$

Squaring both sides:

$$(x+1)^2 + y^2 = 1^2$$

This is the standard equation of a circle:

$$(x - x_0)^2 + (y - y_0)^2 = R^2$$

Comparing the terms, we get centre: $(x_0, y_0) = (-1, 0)$ and radius: $R = 1$.

$$\boxed{\text{B.a circle with } (-1, 0) \text{ as the centre and radius } 1}$$

Q5.94.1 MCQ 1994 1M Ans: C ● ○ ○

Given $z = x + iy$. So, $z^* = x - iy$.

$$z + z^* = (x + iy) + (x - iy) = 2x$$

$$z + z^* = 2\text{Re}(z) \implies \text{Re}(z) = \frac{z + z^*}{2}$$

$$\boxed{C.\ \text{Re}(z) = \frac{z + z^*}{2}}$$

Q5.94.2 MCQ 1994 1M Ans: D ● ○ ○

By Euler's formula, the complex exponential functions are expressed as:

$$e^{i\phi} = \cos\phi + i\sin\phi$$

$$e^{-i\phi} = \cos(-\phi) + i\sin(-\phi) = \cos\phi - i\sin\phi$$

Adding these two equations together:

$$e^{i\phi} + e^{-i\phi} = (\cos\phi + \cos\phi) + (i\sin\phi - i\sin\phi)$$

$$e^{i\phi} + e^{-i\phi} = 2\cos\phi$$

Solving for $\cos\phi$:

$$\cos\phi = \frac{e^{i\phi} + e^{-i\phi}}{2}$$

$$\boxed{D.\ \frac{e^{i\phi} + e^{-i\phi}}{2}}$$

Q5.26.1 MCQ 2026 2M Ans: A ● ○ ○

Let $z = re^{i\theta}$, where $r = \sqrt{x^2 + y^2}$ and $\theta = \tan^{-1}(y/x)$. The complex logarithm function $w = \log_e z$ is defined as:

$$w = \log_e(re^{i\theta}) = \log_e r + i\theta$$

$$w = \frac{1}{2}\log_e(x^2 + y^2) + i\tan^{-1}\left(\frac{y}{x}\right)$$

The principal branch of the complex logarithm is analytic everywhere in the complex plane except at the branch point $z = 0$ and along the negative real axis (the branch cut). Wrt to its domain of definition, it is analytic everywhere except at the singular point $z = 0$.

The harmonic conjugate components are $u = \frac{1}{2}\log_e(x^2 + y^2)$ and $v = \tan^{-1}(y/x)$.

$$\boxed{\text{A. } w \text{ is analytic everywhere except at } z = 0}$$

Q5.25.1 NAT 2025 2M Ans: -0.1 to 0.1 ● ○ ○

Given contour integral be:

$$I = \oint_C \frac{z^3}{(z^2 + 4)(z^2 - 4)}\, dz$$

The contour C is the unit circle centered at the origin, $|z| = 1$. Finding poles,

$$(z^2 + 4)(z^2 - 4) = 0$$

$z^2 - 4 = 0 \implies z = \pm 2$ and $z^2 + 4 = 0 \implies z = \pm 2i$

Since the contour is a unit circle , all these poles lie strictly outside the contour C.

By Cauchy's Integral Theorem, if a function is analytic everywhere inside and on a closed contour C, the loop integral is zero:

$$I = 0 = 2\pi i n => n = 0$$

$$\boxed{0.0}$$

Q5.24.1 MCQ 2024 1M Ans: D ● ● ●

Given the real part of an analytic function $f(z) = u + iv$:

$$u(x, y) = \cosh x \cos y$$

Using Milne-Thomson's method to find $f(z)$:

$$\frac{\partial u}{\partial x} = \sinh x \cos y \implies \phi_1(z, 0) = \sinh z \cos(0) = \sinh z$$

$$\frac{\partial u}{\partial y} = -\cosh x \sin y \implies \phi_2(z, 0) = -\cosh z \sin(0) = 0$$

According to Milne-Thomson's method:

$$f(z) = \int [\phi_1(z, 0) - i\phi_2(z, 0)]\, dz + c$$

$$f(z) = \int \sinh z\, dz + c = \cosh z + c$$

Given that the imaginary part of $f(z)$ is zero for $y = 0$.

Since $\cosh z = \cosh(x + iy) = \cosh x \cos y + i \sinh x \sin y$, at $y = 0$, $\text{Im}\{f(z)\} = \sinh x \sin(0) + \text{Im}\{c\} = 0 \implies \text{Im}\{c\} = 0$.

The simplest solution matching the options with $c = 0$ is:

$$f(z) = \cosh z$$

$$\boxed{\text{D. } \cosh z}$$

Q5.23.1 NAT 2023 1M Ans: 1.999 to 2.001 ● ● ○

Given the function:

$$f(z) = e^{-kx}(\cos 2y - i\sin 2y) = e^{-kx}e^{-i2y} = e^{-(kx+i2y)}$$

Given, $f(z)$ to be a function of $z = x + iy$. If $k = 2$, then:

$$f(z) = e^{-(2x+i2y)} = e^{-2(x+iy)} = e^{-2z}$$

Since e^{-2z} is an exponential function of z, it is an entire (analytic everywhere) function. Thus, the condition for analyticity is satisfied when $k = 2$.

$$\boxed{2}$$

Q5.22.1 MCQ 2022 2M Ans: A ● ○ ○

Given $f(z) = \frac{6z}{2z^4-3z^3+7z^2-3z+5}$. The contour encloses only the pole $z = i$. By the Residue Theorem, the value of the integral is:

$$I = 2\pi i \times \text{Res}(f, z = i)$$

To find the residue at the simple pole $z = i$:

$$\text{Res}(f, z = i) = \lim_{z\to i} \frac{6z}{\frac{d}{dz}\left(2z^4 - 3z^3 + 7z^2 - 3z + 5\right)}$$

$$\text{Res}(f, z = i) = \left.\frac{6z}{8z^3 - 9z^2 + 14z - 3}\right|_{z=i}$$

Substitute $z = i$,

$$\text{Res}(f, z = i) = \frac{6i}{8(-i) - 9(-1) + 14i - 3} = \frac{6i}{-8i + 9 + 14i - 3}$$

$$\text{Res}(f, z = i) = \frac{6i}{6 + 6i} = \frac{i}{1 + i} = \frac{i(1 - i)}{(1 + i)(1 - i)} = \frac{i - i^2}{1 - (-1)} = \frac{1 + i}{2}$$

Finally,

$$I = 2\pi i\left(\frac{1 + i}{2}\right) = \pi i(1 + i) = (-1 + i)\pi$$

$$\boxed{\text{A. } (-1 + i)\,\pi}$$

Q5.22.2 NAT 2022 2M Ans: 0.2 ● ○ ○

The given contour integral is:

$$I = \frac{1}{2\pi}\oint_C \frac{1}{(z - i)(z + 4i)}\,dz$$

The contour C is a circle of radius 2 centered at the origin: $|z| = 2$.
Finding poles,
1. $z_1 = i \implies |z_1| = 1 < 2$ (inside the contour C)
2. $z_2 = -4i \implies |z_2| = 4 > 2$ (outside the contour C)
By Cauchy's Integral Formula, looking only at the enclosed pole $z = i$:

$$\oint_C \frac{\frac{1}{z+4i}}{z - i}\,dz = 2\pi i \times \left.\frac{1}{z + 4i}\right|_{z=i}$$

$$= 2\pi i\left(\frac{1}{i + 4i}\right) = 2\pi i\left(\frac{1}{5i}\right) = \frac{2\pi}{5}$$

Substituting this value back into the expression for I:

$$I = \frac{1}{2\pi}\cdot\frac{2\pi}{5} = \frac{1}{5} = 0.2$$

$$\boxed{0.2}$$

Q5.21.1 MCQ 2021 2M Ans: C ● ● ○

Contour C is the unit circle centered at the origin: $|z| = 1$. To evaluate,

$$I = \oint_C \frac{\cosh 3z}{2z}\,dz = \frac{1}{2}\oint_C \frac{\cosh 3z}{z}\,dz$$

A simple pole at $z = 0$, which lies inside the unit circle $|z| = 1$. Using Cauchy's Integral Formula:

$$\oint_C \frac{f(z)}{z - z_0}\,dz = 2\pi i f(z_0)$$

Here, $f(z) = \cosh 3z$ and $z_0 = 0$:

$$I = \frac{1}{2}\left(2\pi i \cdot \cosh(3 \cdot 0)\right)$$

Since $\cosh(0) = 1$:

$$I = \frac{1}{2}(2\pi i \cdot 1) = \pi i$$

$$\boxed{\text{C } \pi i}$$

Q5.21.2 MCQ 2021 1M Ans: B ● ○ ○

$$1 + i = \sqrt{1^2 + 1^2}e^{i\pi/4} = \sqrt{2}e^{i\pi/4}$$

Therefore,

$$(1+i)^8 = \left(\sqrt{2}e^{i\pi/4}\right)^8$$
$$(1+i)^8 = (\sqrt{2})^8 \cdot e^{i8\pi/4}$$
$$(1+i)^8 = 2^4 \cdot e^{i2\pi}$$

Since $e^{i2\pi} = \cos(2\pi) + i\sin(2\pi) = 1$:

$$(1+i)^8 = 16 \cdot 1 = 16$$

$$\boxed{\text{B. } 16}$$

Q5.20.1 MCQ 2020 1M Ans: D ● ○ ○

An analytic function is a function that is differentiable at every point in its domain.

$f(z) = z^2$, $f(z) = e^z$, and $f(z) = \sin z$ are polynomials, exponential, and trigonometric combinations that are well-defined and differentiable over the entire complex plane (entire functions).

$f(z) = \log z$ is not defined at $z = 0$, and has a branch point at the origin. Thus, it is NOT analytic at all points of the complex plane.

$$\boxed{\text{D. } f(z) = log\ z}$$

Q5.20.2 NAT 2020 2M Ans: 1.99 to 2.01 ● ● ○

Given the analytic function $f(z) = x^2 - y^2 + i\psi(x,y)$, where $u(x,y) = x^2 - y^2$ and $v(x,y) = \psi(x,y)$. Using the Cauchy-Riemann equations:

$$\frac{\partial \psi}{\partial y} = \frac{\partial u}{\partial x} = 2x \implies \psi = 2xy + g(x)$$
$$\frac{\partial \psi}{\partial x} = -\frac{\partial u}{\partial y} = -(-2y) = 2y$$

Differentiating our expression for ψ with respect to x:

$$2y + g'(x) = 2y \implies g'(x) = 0 \implies g(x) = c$$

Assuming $c = 0$, the imaginary part is:

$$\psi(x, y) = 2xy$$

At $z = 1 + i$, $i.e.$, $x = 1$ and $y = 1$:

$$\psi(1, 1) = 2(1)(1) = 2$$

$$\boxed{2}$$

Q5.20.3 MCQ | 2020 | 2M | Ans: D | ● ● ●

Given $f(z) = u + iv$ where the real part is $u(x, y) = x^3 - 3xy^2$. Using the Cauchy-Riemann equations to determine $v(x, y)$:

$$\frac{\partial u}{\partial x} = 3x^2 - 3y^2$$

$$\frac{\partial u}{\partial y} = -6xy$$

From the relations:

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = 3x^2 - 3y^2$$

Integrating with respect to y:

$$v(x, y) = 3x^2y - y^3 + g(x)$$

Differentiating with respect to x:

$$\frac{\partial v}{\partial x} = 6xy + g'(x)$$

Equating to $-\frac{\partial u}{\partial y}$:

$$6xy + g'(x) = -(-6xy) = 6xy \implies g'(x) = 0 \implies g(x) = \text{constant}$$

Thus,

$$v(x, y) = (3x^2y - y^3) + \text{constant}$$

$$\boxed{D.(3x^2y - y^3) + \text{constant}}$$

Q5.19.1 MCQ | 2019 | 1M | Ans: B | ● ○ ○

For a complex function $f(z) = u(x, y) + iv(x, y)$ to be analytic in a domain, it must satisfy the standard Cauchy-Riemann equations at all points in that domain:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\boxed{\text{B. } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}}$$

Q5.18.1 MCQ 2018 2M Ans: A ● ○ ○

Given the contour integral:

$$\oint_C \frac{1}{5z-4}\,dz = \oint_C \frac{1}{5\left(z-\frac{4}{5}\right)}\,dz = \frac{1}{5}\oint_C \frac{1}{z-\frac{4}{5}}\,dz$$

The contour C is a unit circle centered at the origin, $|z| = 1$. Simple pole exists at $z_0 = \frac{4}{5} = 0.8$. Since $|0.8| < 1$, the pole lies completely inside the contour C. Applying Cauchy's Integral Formula:

$$\oint_C \frac{1}{z-z_0}\,dz = 2\pi i$$

Therefore,

$$\frac{1}{5}(2\pi i) = \frac{2}{5}\pi i$$

$$A\pi i = \frac{2}{5}\pi i \implies A = \frac{2}{5}$$

$$\boxed{A.\frac{2}{5}}$$

Q5.17.1 MCQ 2017 2M Ans: B ● ○ ○

Given $f(z) = (x^2 + ay^2) + ibxy$, so $u = x^2 + ay^2$ and $v = bxy$.
Applying the Cauchy-Riemann equations:
1) $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$:

$$2x = bx \implies b = 2$$

2) $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$:

$$2ay = -by = -2y \implies a = -1$$

Thus, $a = -1, b = 2$.

$$\boxed{\text{B. a=-1 , b=2}}$$

Q5.16.1 MCQ 2016 1M Ans: A ● ● ○

Given $u(x, y) = 2xy$.

$$\frac{\partial u}{\partial x} = 2y, \quad \frac{\partial u}{\partial y} = 2x$$

Using Cauchy-Riemann equations:

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = 2y \implies v = y^2 + g(x)$$

$$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} \implies g'(x) = -2x \implies g(x) = -x^2 + \text{constant}$$

Thus, $v(x, y) = -x^2 + y^2 + \text{constant}$.

$$\boxed{\text{A. } -x^2 + y^2 + \text{constant}}$$

Q5.16.2 NAT 2016 1M Ans: -1.1 to -0.9 ● ● ○

Given $u(x, y) = 2kxy$ and $v(x, y) = x^2 - y^2$. Using Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = 2ky, \quad \frac{\partial v}{\partial y} = -2y$$

$$\frac{\partial u}{\partial y} = 2kx, \quad \frac{\partial v}{\partial x} = 2x$$

From the first condition of the CR equations ($\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$):

$$2ky = -2y \implies k = -1$$

From the second condition of CR equations ($\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$):

$$2(-1)x = -2x \implies -2x = -2x \quad \text{(Verified)}$$

$$\boxed{-1}$$

Q5.16.3 MCQ 2016 2M Ans: B ● ● ●

Let the given integral be:

$$\oint_{\Gamma} \frac{3z - 5}{(z-1)(z-2)}\, dz$$

Using partial fractions,

$$\frac{3z-5}{(z-1)(z-2)} = \frac{\alpha}{z-1} + \frac{\beta}{z-2}$$

$$\alpha = \lim_{z\to 1} \frac{3z-5}{z-2} = \frac{-2}{-1} = 2$$

$$\beta = \lim_{z\to 2} \frac{3z-5}{z-1} = \frac{1}{1} = 1$$

So the integral becomes:

$$\oint_{\Gamma} \left(\frac{2}{z-1} + \frac{1}{z-2} \right) dz$$

By Residue Theorem, a counter-clockwise loop around a pole z_0 contributes $2\pi i \cdot$ Residue.

If Γ encloses only $z = 1$: Value $= 2(2\pi i) = 4\pi i$.
If Γ encloses only $z = 2$: Value $= 1(2\pi i) = 2\pi i$.
If Γ encloses both poles: Value $= (2 + 1)2\pi i = 6\pi i$.

Since the value is given as $4\pi i$, the path Γ must enclose **only** the singularity at $z = 1$ in a counter-clockwise direction. Looking at the options, diagram B depicts a counter-clockwise circle enclosing $z = 1$ and excluding $z = 2$.

$$\boxed{\text{B}}$$

Q5.15.1 MCQ 2015 1M Ans: A ● ○ ○

Given complex numbers:

$$z_1 = 5 + 5\sqrt{3}i \implies \theta_1 = \tan^{-1}\left(\frac{5\sqrt{3}}{5}\right) = \tan^{-1}(\sqrt{3}) = 60^\circ$$

$$z_2 = \frac{2}{\sqrt{3}} + 2i \implies \theta_2 = \tan^{-1}\left(\frac{2}{2/\sqrt{3}}\right) = \tan^{-1}(\sqrt{3}) = 60^\circ$$

Using the property of arguments:

$$\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$$

$$\arg\left(\frac{z_1}{z_2}\right) = 60° - 60° = 0°$$

$$\boxed{\text{A. } 0}$$

Q5.14.1 MCQ 2014 1M Ans: C ● ○ ○

Given complex number is $z = \frac{1+i}{1-i}$. Simplifying,

$$z = \frac{(1+i)(1+i)}{(1-i)(1+i)} = \frac{1+2i+i^2}{1-i^2} = \frac{1+2i-1}{1-(-1)} = \frac{2i}{2} = i$$

The number $z = i$ lies on the positive imaginary axis, so its argument is:

$$\arg(i) = \frac{\pi}{2}$$

$$\boxed{\text{C. } \pi/2}$$

Q5.14.2 MCQ 2014 2M Ans: C ● ● ●

Given the real part of an analytic function:

$$u(x, y) = x^2 - y^2$$

For $f(z) = u + iv$ to be analytic, it must satisfy the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Differentiating $u(x, y)$ with respect to x and y:

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial u}{\partial y} = -2y$$

From the first Cauchy-Riemann condition:

$$\frac{\partial v}{\partial y} = 2x$$

Integrating both sides with respect to y:

$$v(x, y) = 2xy + g(x)$$

Differentiating $v(x, y)$ with respect to x:

$$\frac{\partial v}{\partial x} = 2y + g'(x)$$

Using the second Cauchy-Riemann condition ($\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$):

$$2y + g'(x) = -(-2y) \implies 2y + g'(x) = 2y \implies g'(x) = 0$$

Integrating $g'(x)$ yields a constant c:

$$g(x) = c$$

Thus, the imaginary part expression is:

$$v(x, y) = 2xy + c$$

$$\boxed{\text{C. } 2xy + c}$$

Q5.11.1 MCQ | 2011 | 1M | Ans: A | ● ○ ○

Given $z_1 = 1 + i$ and $z_2 = 2 - 5i$:

$$z_1 z_2 = (1+i)(2-5i) = 1(2) - 1(5i) + i(2) - i(5i) = 2 - 5i + 2i - 5i^2 = 2 - 3i + 5 = 7 - 3i$$

$$\boxed{\text{A. 7-3}i}$$

Q5.10.1 MCQ | 2010 | 1M | Ans: B | ● ○ ○

Given complex number fraction is:

$$\frac{3+4i}{1-2i} = \frac{(3+4i)(1+2i)}{(1-2i)(1+2i)} = \frac{-5+10i}{5} = -1 + 2i$$

Taking the modulus:

$$\therefore \left|\frac{3+4i}{1-2i}\right| = |-1+2i| = \sqrt{5}$$

$$\boxed{\text{B. } \sqrt{5}}$$

Key Points

- The modulus of a division of complex numbers satisfies $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$.
- For a complex number $z = x + iy$, the modulus is $|z| = \sqrt{x^2 + y^2}$.

Q5.09.1 MCQ | 2009 | 2M | Ans: C | ● ● ●

Here u and v are analytic as $f(z)$ is analytic. Therefore, u, v satisfy the Cauchy-Riemann equations:

$$u_x = v_y \quad \text{—(i)} \quad \text{and} \quad u_y = -v_x \quad \text{—(ii)}$$

Given:

$$u = xy$$

Differentiating u with respect to x:

$$\therefore u_x = y$$

From equation (i):

$$\Rightarrow v_y = y$$

Integrating with respect to y:

$$v = \frac{y^2}{2} + c(x) \quad \text{—(iii)}$$

Differentiating v with respect to x:

$$v_x = c'(x)$$

From equation (ii):

$$\Rightarrow -u_y = c'(x)$$

Since $u = xy \Rightarrow u_y = x$, we have:

$$\Rightarrow -x = c'(x)$$

Integrating with respect to x:

$$c(x) = \frac{-x^2}{2} + k$$

Substituting $c(x)$ into equation (iii), we get:

$$v = \frac{y^2 - x^2}{2} + k$$

$$\boxed{\text{C. } \frac{y^2 - x^2}{2} + k}$$

Key Points

- For an analytic function $f(z) = u + iv$, the real and imaginary parts must satisfy the Cauchy-Riemann equations: $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$.

Q5.08.1 MCQ 2008 2M Ans: A ● ● ○

Given function is:

$$f(z) = \frac{\cos z}{z}$$

Here, $f(z)$ has a simple pole at $z = 0$. The residue at $z = 0$ is given by:

$$\text{Res}(f, 0) = \lim_{z\to 0} [(z - 0)f(z)] = 1$$

By Cauchy's Residue Theorem, the integral evaluated around the unit circle is:

$$\oint f(z)\,dz = 2\pi i \times 1 = 2\pi i$$

$$\boxed{\text{A. } 2\pi i}$$

Key Points

- According to Cauchy's Residue Theorem, if a function $f(z)$ is analytic within and on a closed contour C except at a finite number of poles inside C, then $\oint_C f(z)\,dz = 2\pi i \sum \text{Res}$.
- The residue at a simple pole $z = a$ is calculated as $\lim_{z\to a}(z - a)f(z)$.

Q5.07.1 MCQ 2007 1M Ans: B ● ○ ○

$\Phi(x, y) + i\Psi(x, y)$ is analytic, so it satisfies Cauchy-Riemann equation

$$\frac{\partial \Phi}{\partial x} = \frac{\partial \Psi}{\partial y}, \quad \frac{\partial \Phi}{\partial y} = -\frac{\partial \Psi}{\partial x}$$

$$\boxed{\text{B. } \frac{\partial \Phi}{\partial x} = \frac{\partial \Psi}{\partial y}; \quad \frac{\partial \Phi}{\partial y} = -\frac{\partial \Psi}{\partial x}}$$

Key Points

- For any complex function $f(z) = u + iv$ to be analytic, the component functions must satisfy the standard Cauchy-Riemann equations: $u_x = v_y$ and $u_y = -v_x$.

Q5.06.1 MCQ 2006 2M Ans: A ● ○ ○

The given integral is:

$$\int_0^{\pi/3} e^{it}\,dt$$

Evaluating the integral:

$$\int_0^{\pi/3} e^{it}\,dt = \left[\frac{e^{it}}{i}\right]_0^{\pi/3} = \frac{1}{i}\left(e^{i\pi/3} - 1\right) = -i\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3} - 1\right)$$

Substituting the trigonometric values:

$$= -i\left(\frac{1}{2} + i\frac{\sqrt{3}}{2} - 1\right) = \frac{\sqrt{3}}{2} + i\frac{1}{2}$$

$$\boxed{\text{A. } \frac{\sqrt{3}}{2} + i\frac{1}{2}}$$

Q5.11.1 MCQ 2011 1M Ans: D ● ● ○

Cauchy Riemann equations for

$$f(z) = u + iv$$

$$u_x = v_y \text{ and } u_y = -v_x \quad \ldots \text{(ii)}$$

Given $u = 3x^2 - 3y^2$

$$\Rightarrow u_x = 6x \Rightarrow u_y = 6x \quad \text{[using (i)]}$$

$$\Rightarrow v = 6xy + \phi(x) \quad \ldots \text{(iii)}$$

Differentiating w.r.t. x

$$v_x = 6y + \phi'(x)$$

$$\Rightarrow -u_y = 6y + \phi'(x)$$

$$\Rightarrow 6y = 6y + \phi'(x) \quad \text{[using (ii)]}$$

$$\Rightarrow \phi'(x) = 0$$

$$\Rightarrow \phi(x) = K \text{ (Constant)}$$

$\therefore$ from (iii) we get

$$v = 6xy + K$$

$$\boxed{\text{D. } 6xy + K}$$

Q5.10.1 MCQ 2010 1M Ans: D ● ● ○

Let the given improper integral be:

$$I = \int_{-\infty}^{\infty} \frac{dx}{1 + x^2}$$

Since the integrand $f(x) = \frac{1}{1+x^2}$ is an even function, we can rewrite the limits of integration:

$$I = 2 \int_{0}^{\infty} \frac{dx}{1 + x^2}$$

The integral expression $\int \frac{1}{1+x^2}\, dx = \tan^{-1}(x) + C$ is a fundamental calculus identity.

$$I = 2 \left[\tan^{-1}(x)\right]_0^{\infty}$$

Applying the upper and lower limits:

$$I = 2 \left(\lim_{x \to \infty} \tan^{-1}(x) - \tan^{-1}(0) \right)$$

$$I = 2 \left(\frac{\pi}{2} - 0 \right) = \pi$$

$$\boxed{\text{D. } \pi}$$

Q5.09.1 MCQ 2009 1M Ans: D ● ○ ○

$$f(z) = \frac{z - 1}{z^2 + 1} = \frac{z - 1}{(z + i)(z - i)}$$

$\therefore$ $f(z)$ has singularities at $z = i, -i$

$$\boxed{\text{D. i and -i}}$$

Q5.09.2 MCQ 2009 2M Ans: C ● ○ ○

Let $f(z) = \frac{\cos(2\pi z)}{(2z-1)(z-3)}$. $f(z)$ has singularity at $z = 1/2$ in C ($|z| = 1$).

$$\therefore \text{Res}(f, 1/2) = \lim_{z\to 1/2} [(z - 1/2) f(z)]$$

$$= \lim_{z\to 1/2} \left[\frac{1}{2} \cdot \frac{\cos(2\pi z)}{z-3}\right]$$

$$= \frac{1}{5}$$

$$\therefore \int_c f(z)\, dz = \frac{2\pi i}{5}$$

$$\boxed{\text{C. } 2\pi i/5}$$

Q5.07.1 MCQ 2007 2M Ans: A ● ● ●

Given the velocity potential function:

$$\phi = x^2 - y^2$$

For an irrotational, incompressible fluid flow, the relation between the velocity components, the potential function ϕ, and the stream function ψ is given by:

$$u = -\frac{\partial\phi}{\partial x} = -\frac{\partial\psi}{\partial y}$$

$$v = -\frac{\partial\phi}{\partial y} = \frac{\partial\psi}{\partial x}$$

Finding the partial derivatives of ϕ:

$$\frac{\partial\phi}{\partial x} = 2x$$

$$\frac{\partial\phi}{\partial y} = -2y$$

Using the first relation to find ψ:

$$-\frac{\partial\psi}{\partial y} = 2x \implies \frac{\partial\psi}{\partial y} = -2x$$

Integrating with respect to y:

$$\psi = -2xy + f(x)$$

Differentiate this expression for ψ with respect to x:

$$\frac{\partial\psi}{\partial x} = -2y + f'(x)$$

Using the second relation where $\frac{\partial\psi}{\partial x} = -\frac{\partial\phi}{\partial y}$:

$$-2y + f'(x) = -(-2y) = 2y$$

$$f'(x) = 4y$$

Note: The standard sign convention in fluid mechanics can vary, but alternatively using standard Cauchy-Riemann related equations for complex potential $w = \phi + i\psi$ where ϕ and ψ are harmonic conjugates:

$$\frac{\partial\phi}{\partial x} = \frac{\partial\psi}{\partial y} \implies 2x = \frac{\partial\psi}{\partial y} \implies \psi = 2xy + f(x)$$

$$\frac{\partial\phi}{\partial y} = -\frac{\partial\psi}{\partial x} \implies -2y = -(2y + f'(x)) \implies f'(x) = 0 \implies f(x) = C$$

Thus, the stream function is:

$$\psi = 2xy + C$$

Applying the boundary condition $\psi = 0$ at $x = 0, y = 0$:

$$0 = 2(0)(0) + C \implies C = 0$$

Therefore, the stream function is:

$$\psi = 2xy$$

$$\boxed{\text{A. 2xy}}$$

Q5.06.1 MCQ 2006 1M Ans: A ● ● ○

Let $f(z) = \frac{z^3-6}{3z-i}$. Here $f(z)$ has a singularities at $z = i/3$

$$\therefore \text{Res}(f, \frac{1}{3}) = \lim_{z \to i/3}\left[(z - i/3) \times \frac{z^3 - 6}{3z - i}\right] = \frac{1}{3} \cdot \left(\frac{i^3}{27} - 6\right)$$

$$\therefore \int_c f(z)\,dx = 2\pi i \times \frac{1}{3}\left(\frac{i^3}{27} - 6\right) = \frac{2\pi}{81} i^4 - 4\pi i = \frac{2\pi}{81} - 4\pi i$$

$$\boxed{\text{A. } \frac{2\pi}{81} - 4\pi i}$$

Mistakes to be avoided

Don't forget to take common such that the pole is of form (z-a) in denominator before cancelling.

Q5.05.1 MCQ 2005 1M Ans: C ● ● ●

(a) is true since

$$\frac{Z_1}{Z_2} = \frac{Z_1\bar{Z}_2}{Z_2\bar{Z}_2} = \frac{Z_1\bar{Z}_2}{|Z_2|^2}$$

(b) is true by triangle inequality of complex number.
(c) is not true since $|Z_1 - Z_2| \geq |Z_1| - |Z_2|$
(d) is true since

$$|Z_1 + Z_2|^2 = (Z_1 + Z_2)\overline{(Z_1 + Z_2)}$$
$$= (Z_1 + Z_2)(\bar{Z}_1 + \bar{Z}_2)$$
$$= Z_1\bar{Z}_1 + Z_2\bar{Z}_2 + Z_2\bar{Z}_1 + Z_1\bar{Z}_2 \quad \ldots \text{(i)}$$

And

$$|Z_1 - Z_2|^2 = (Z_1 - Z_2)(\bar{Z}_1 - \bar{Z}_2)$$
$$= Z_1\bar{Z}_1 + Z_2\bar{Z}_2 - Z_2\bar{Z}_1 - Z_1\bar{Z}_2 \quad \ldots \text{(ii)}$$

Adding (i) and (ii) we get

$$|Z_1 + Z_2|^2 + |Z_1 - Z_2|^2 = 2Z_1\bar{Z}_1 + 2Z_2\bar{Z}_2$$
$$= 2|Z_1|^2 + 2|Z_2|^2$$

$$\boxed{\text{C. } |Z_1 - Z_2| \leq |Z_1| - |Z_2|}$$

Key Points

- The triangle inequalities state that $|z_1 + z_2| \leq |z_1| + |z_2|$ and $|z_1 - z_2| \geq ||z_1| - |z_2||$.
- The parallelogram law for complex numbers dictates that $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$.

Q5.05.2 MCQ 2005 2M Ans: A ● ● ●

$$\int \sec z\, dz = \int \frac{1}{\cos z}\, dz$$

The poles are at $z_0 = \left(n + \frac{1}{2}\right)\pi$

$$= \ldots\ldots \frac{-3\pi}{2}, \frac{-\pi}{2}, \frac{\pi}{2}, \frac{+3\pi}{2} \ldots\ldots$$

None of these poles lie inside the unit circle $|z| = 1$ Hence, sum of residues at poles $= 0$ $\therefore$ Singularities set $= \phi$ and

$$\text{I} = 2\pi i \text{ [sum of residues of f(z) at the poles]}$$

$$= 2\pi i \times 0 = 0$$

$$\boxed{\text{A. } I = 0 : \text{ singularities set} = \phi}$$

Key Points

The unit circle $|z| = 1$ has a radius of 1, whereas the closest poles of $\sec z$ occur at $\pm\pi/2 \approx \pm 1.57$, which lie entirely outside the contour.

Q5.26.1 MCQ 2026 2M Ans: D ● ● ○

Given two complex numbers,

$$z_1 = r_1(\cos\theta_1 + i\sin\theta_1) = r_1 e^{i\theta_1}$$
$$z_2 = r_2(\cos\theta_2 + i\sin\theta_2) = r_2 e^{i\theta_2}$$

where $r_1, r_2 \geq 0$ are their respective magnitudes ($|z_1| = r_1$ and $|z_2| = r_2$). Here, the triangle inequality becomes an equality:

$$|z_1 + z_2| = |z_1| + |z_2|$$
$$z_1 + z_2 = (r_1\cos\theta_1 + r_2\cos\theta_2) + i(r_1\sin\theta_1 + r_2\sin\theta_2)$$

Manipulating the above two equations and expanding,

$$|z_1 + z_2|^2 = (r_1\cos\theta_1 + r_2\cos\theta_2)^2 + (r_1\sin\theta_1 + r_2\sin\theta_2)^2$$
$$|z_1 + z_2|^2 = \left(r_1^2\cos^2\theta_1 + r_2^2\cos^2\theta_2 + 2r_1r_2\cos\theta_1\cos\theta_2\right) + \left(r_1^2\sin^2\theta_1 + r_2^2\sin^2\theta_2 + 2r_1r_2\sin\theta_1\sin\theta_2\right)$$
$$|z_1 + z_2|^2 = r_1^2(\cos^2\theta_1 + \sin^2\theta_1) + r_2^2(\cos^2\theta_2 + \sin^2\theta_2) + 2r_1r_2(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2)$$

Since $\cos^2\theta + \sin^2\theta = 1$ and $\cos(A - B) = \cos A\cos B + \sin A\sin B$:

$$|z_1 + z_2|^2 = r_1^2 + r_2^2 + 2r_1r_2\cos(\theta_1 - \theta_2) \quad \text{— (Identity 1)}$$

Squaring the RHS of given equation,

$$(|z_1| + |z_2|)^2 = (r_1 + r_2)^2 = r_1^2 + r_2^2 + 2r_1r_2 \quad \text{— (Identity 2)}$$

Equating Identity 1 and Identity 2 as given by our condition $|z_1 + z_2|^2 = (|z_1| + |z_2|)^2$:

$$r_1^2 + r_2^2 + 2r_1r_2\cos(\theta_1 - \theta_2) = r_1^2 + r_2^2 + 2r_1r_2$$
$$2r_1r_2\cos(\theta_1 - \theta_2) = 2r_1r_2$$

Assuming non-trivial complex numbers where $r_1 \neq 0$ and $r_2 \neq 0$, we can divide by $2r_1r_2$:

$$\cos(\theta_1 - \theta_2) = 1$$

For this to hold, the argument must be an integer multiple of 2π:

$$\theta_1 - \theta_2 = 2k\pi \quad \text{for } k \in \mathbb{Z}$$

Given $0 \leq \theta_1 \leq \frac{\pi}{2}$ and $0 \leq \theta_2 \leq \frac{\pi}{2}$, the maximum possible difference between the two arguments is bounded within $[-\frac{\pi}{2}, \frac{\pi}{2}]$. Therefore, the only valid integer solution is $k = 0$:

$$\theta_1 - \theta_2 = 0 \implies \theta_1 = \theta_2$$

$$\boxed{\text{D. } \theta_1 = \theta_2}$$

Key Points

- Geometrically, the equality condition $|z_1 + z_2| = |z_1| + |z_2|$ states that the length of the combined vector equals the sum of the individual vector lengths, meaning the two vectors are perfectly collinear and point in identical directions.

Q5.25.1 MCQ 2025 1M Ans: B ● ○ ○

Method 1:Using Property

Given $z_1 = 1 - i$ and $z_2 = i$.

$$\text{Arg}(z_1) = \tan^{-1}\left(\frac{-1}{1}\right) = -\frac{\pi}{4}$$

$$\text{Arg}(z_2) = \frac{\pi}{2}$$

Using the property of complex arguments:

$$\text{Arg}(z_1 z_2) = \text{Arg}(z_1) + \text{Arg}(z_2)$$

$$= -\frac{\pi}{4} + \frac{\pi}{2} = \frac{\pi}{4}$$

Method 2:Direct

$$\text{z}_1 z_2 = (1 - i).i = (\text{i} + 1)$$

Therefore,

$$\text{Arg}(z_1 z_2) = \text{Arg}(1 + i) = \frac{\pi}{4}$$

$$\boxed{\text{B. } \pi/4}$$

Q5.24.1 MCQ 2024 1M Ans: B ● ○ ○

Given $z_1 = -1 + i$ and $z_2 = 2i$.

$$\text{Arg}(z_1) = \pi - \tan^{-1}\left(\frac{1}{1}\right) = \pi - \frac{\pi}{4} = \frac{3\pi}{4} \quad (\text{since } z_1 \text{ is in quadrant II})$$

$$\text{Arg}(z_2) = \frac{\pi}{2} \quad (\text{purely imaginary positive number})$$

Using the property of complex arguments for quotients:

$$\text{Arg}\left(\frac{z_1}{z_2}\right) = \text{Arg}(z_1) - \text{Arg}(z_2)$$

$$= \frac{3\pi}{4} - \frac{\pi}{2} = \frac{\pi}{4}$$

$$\boxed{\text{B. } \pi/4}$$

Q5.23.1 MCQ 2023 1M Ans: B ● ○ ○

Given $z_1 = 2 + 3i$ and $z_2 = -2 + 3i$.

$$|z_1| = \sqrt{2^2 + 3^2} = \sqrt{13}$$

$$|z_2| = \sqrt{(-2)^2 + 3^2} = \sqrt{13}$$

$$z_1 + z_2 = (2 - 2) + (3i + 3i) = 6i$$

$$|z_1 + z_2| = |6i| = 6$$

$$F = \frac{|z_1 + z_2|}{|z_1| + |z_2|} = \frac{6}{\sqrt{13} + \sqrt{13}} = \frac{6}{2\sqrt{13}} = \frac{3}{\sqrt{13}}$$

Since $\sqrt{13} > 3$, it follows that $F < 1$. Also, since it is a ratio of magnitudes, $F > 0$.

$$\boxed{\text{B. F<1}}$$

Key Points

- Interpretation-By the triangle inequality, $|z_1 + z_2| < |z_1| + |z_2|$ whenever z_1 and z_2 are not collinear along the same ray, yielding a ratio $F < 1$.

Q5.20.1 MCQ 2020 1M Ans: C ● ● ○

P. $\tanh \mathbf{x} = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} \implies \mathbf{IV}$
Q. $\coth \mathbf{x} = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}} \implies \mathbf{I}$
R. $\text{sech}\, \mathbf{x} = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}} \implies \mathbf{II}$
S. $\text{cosech}\, \mathbf{x} = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}} \implies \mathbf{III}$

$$\boxed{\text{C. P–IV, Q–I, R–II, S–III}}$$

Q5.19.1 MCQ 2019 2M Ans: A ● ○ ○

We need to find the value of $i^{-1/2} = \frac{1}{\sqrt{i}}$. Express i in exponential form:

$$i = e^{i\frac{\pi}{2}}$$

Therefore, taking the square root:

$$i^{1/2} = e^{i\frac{\pi}{4}} = \cos\frac{\pi}{4} + i\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}(1+i)$$

Now find the reciprocal $i^{-1/2}$:

$$i^{-1/2} = \frac{1}{e^{i\frac{\pi}{4}}} = e^{-i\frac{\pi}{4}} = \cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)$$

$$= \cos\frac{\pi}{4} - i\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}(1-i)$$

$$\boxed{\text{A. } \frac{1}{\sqrt{2}}(1-i)}$$

Q5.17.1 NAT 2017 1M Ans: 2.9 to 3.1 ● ○ ○

The given complex number is:

$$z = 6e^{i\pi/3}$$

Using Euler's formula,

$$z = 6\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$z = 6\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 3 + i3\sqrt{3}$$

The real part of the expression is 3.

$$\boxed{3}$$

Q5.16.1 MCQ 2016 1M Ans: D ● ○ ○

Given complex number is $z = 3 + 4i$. Calculate modulus r ,

$$r = |z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$$

Since the complex number lies in the first quadrant (both real and imaginary parts are positive), the principal argument θ is:

$$\theta = \tan^{-1}\left(\frac{4}{3}\right)$$

$$\boxed{\text{D. } r = 5, \quad \theta = \tan^{-1}\left(\frac{4}{3}\right)}$$

Q5.15.1 MCQ 2015 1M Ans: A ● ● ●

For an analytic function $f(z) = u+iv$, the complex derivative can be expressed in terms of its partial derivatives,

$$\frac{df}{dz} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}$$

Using the Cauchy-Riemann equations ($\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$), we can substitute the terms:
Substituting $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ into the real part yields: $\frac{df}{dz} = \frac{\partial v}{\partial y} + i\frac{\partial v}{\partial x}$ (Option D)
Substituting $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$ into the imaginary part yields: $\frac{df}{dz} = \frac{\partial u}{\partial x} - i\frac{\partial u}{\partial y}$ (Option B)
Differentiation with respect to y gives: $\frac{df}{dz} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$ (Option C)
Evaluating Option A, $\frac{\partial v}{\partial y} + i\frac{\partial u}{\partial y}$, doesn't match anything.

$$\boxed{\text{A. } \frac{df}{dz} = \frac{\partial v}{\partial y} + i\frac{\partial u}{\partial y}}$$

Q5.15.2 NAT 2015 2M Ans: 0.31 to 0.35 ● ● ○

The given integral is:

$$\frac{1}{2\pi i}\oint_C \frac{e^{-2z}}{z(z-3)}\,dz$$

Simple poles exist at $z = 0$ and $z = 3$.
The contour C is defined by $|z| = 2$ - A circle centered at the origin with a radius of 2.
Pole $z = 0$ lies inside the contour ($|0| < 2$), while pole $z = 3$ lies outside the contour ($|3| > 2$).
Using Cauchy's Integral Formula for the internal pole at $a = 0$, where $f(z) = \frac{e^{-2z}}{z-3}$:

$$\oint_C \frac{f(z)}{z-0}\,dz = 2\pi i \cdot f(0)$$

Evaluating $f(0)$:

$$f(0) = \frac{e^{-2(0)}}{0-3} = -\frac{1}{3} \approx -0.33$$

Due to the clockwise direction of the contour C, the integral changes sign:

$$\text{Value} = -\left[\frac{1}{2\pi i} \cdot 2\pi i \cdot f(0)\right] = -f(0) = -\left(-\frac{1}{3}\right) = \frac{1}{3} \approx 0.33$$

$$\boxed{0.33}$$

Mistakes to be avoided

Do not forget that Clockwise contour integration introduces a negative sign factor.

Q5.14.1 MCQ 2014 1M Ans: C ● ○ ○

Given $f(x) = \cos(x) + i\sin(x) = e^{ix}$. Complex conjugate $f^*(x) = \cos(x) - i\sin(x) = e^{-ix}$. Evaluating the integrand product:

$$f^*(x)f(x) = e^{-ix} \cdot e^{ix} = e^0 = 1$$

Therefore,

$$\int_a^b f^*(x)f(x)\,dx = \int_a^b 1\,dx = b - a$$

Since a and b are real numbers, the result $(b-a)$ is real.

$$\boxed{\text{C. Real}}$$

Key Points

Interpretation - The product of any complex function with its own complex conjugate is the square of its absolute magnitude, $|f(x)|^2$, which is always a real-valued function.

Q5.12.1 MCQ 2012 2M Ans: B ● ○ ○

The given integral is:

$$\oint_C \frac{7z+i}{z(z^2+1)}\,dz = \oint_C \frac{7z+i}{z(z+i)(z-i)}\,dz$$

The singularities occur at $z = 0, i, -i$. The contour $|z| < 2$ is a circle of radius 2 centered at the origin, meaning all three poles lie inside C. Finding residues at each singularity,

$$\text{Res}(0) = \left.\frac{7z+i}{z^2+1}\right|_{z=0} = \frac{i}{1} = i$$

$$\text{Res}(i) = \left.\frac{7z+i}{z(z+i)}\right|_{z=i} = \frac{7i+i}{i(2i)} = \frac{8i}{-2} = -4i$$

$$\text{Res}(-i) = \left.\frac{7z+i}{z(z-i)}\right|_{z=-i} = \frac{-7i+i}{-i(-2i)} = \frac{-6i}{-2} = 3i$$

Sum of residues:

$$\sum \text{Res} = i - 4i + 3i = 0$$

By Cauchy's Residue Theorem, the integral value is $2\pi i \times 0 = 0$.

$$\boxed{\text{B. } 0}$$

Q5.08.1 MCQ 2008 2M Ans: C ● ● ●

Given the real part $u(x,y)$ of an analytic function $w(z) = u + iv$:

$$u = \frac{y}{x^2+y^2}$$

We know,

$$x = \frac{z+\bar{z}}{2}, \quad y = \frac{z-\bar{z}}{2i}, \quad x^2+y^2 = |z|^2 = z\bar{z}$$

Substituting into the expression for u and simplifying,

$$u = \frac{\frac{z-\bar{z}}{2i}}{z\bar{z}} = \frac{1}{2i}\left(\frac{z}{z\bar{z}} - \frac{\bar{z}}{z\bar{z}}\right) = \frac{1}{2i}\left(\frac{1}{\bar{z}} - \frac{1}{z}\right)$$

$$u = -\frac{i}{2}\left(\frac{1}{\bar{z}} - \frac{1}{z}\right) = \frac{i}{2z} - \frac{i}{2\bar{z}}$$

For any complex number, $\xi + \bar{\xi} = 2\mathrm{Re}(\xi)$.

$$u = \frac{i}{2z} + \overline{\left(\frac{i}{2z}\right)} = \mathrm{Re}\left(\frac{i}{z}\right)$$

Since $w(z)$ is analytic, so $\mathrm{Re}[w(z)] = u$. By direct comparison, we find:

$$w(z) = \frac{i}{z}$$

$$\boxed{\text{C. } \frac{i}{z}}$$

Q5.07.1 MCQ 2007 2M Ans: D ● ● ●

Given function be $f(z) = z\bar{z}$.

$$f(z) = (x+iy)(x-iy) = x^2 + y^2$$

Thus, the real part $u(x, y) = x^2 + y^2$ and the imaginary part $v(x, y) = 0$. To check the differentiability of $f(z)$, we test the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Computing the partial derivatives:

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial u}{\partial y} = 2y$$
$$\frac{\partial v}{\partial x} = 0, \quad \frac{\partial v}{\partial y} = 0$$

Equating them according to the Cauchy-Riemann conditions:

$$2x = 0 \implies x = 0$$

$$2y = 0 \implies y = 0$$

The Cauchy-Riemann equations are satisfied only at the origin $(0, 0)$, which means $f(z) = z\bar{z}$ is differentiable exclusively at $z = 0$. Since the question asks for the derivative at $z = 2 + i$, where the Cauchy-Riemann equations are violated ($4 \neq 0$ and $2 \neq 0$), the derivative does not exist at this point.

$$\boxed{\text{D. Does not exist}}$$

Key Points

A complex function $f(z) = u + iv$ is differentiable at a point only if it satisfies the Cauchy-Riemann equations: $u_x = v_y$ and $u_y = -v_x$. The function $f(z) = |z|^2 = z\bar{z}$ is continuous everywhere but differentiable nowhere except at the origin $z = 0$.

Mistakes to be avoided

- Do not treat $\bar{z}$ as a constant and apply standard real-variable power rules like $\frac{d}{dz}(z\bar{z}) = \bar{z}$. In complex analysis, $\bar{z}$ is not an analytic function of z.

Q5.26.1 MCQ 2026 1M Ans: A ● ○ ○

Given expression is:

$$\frac{1}{i^n}$$

where n is an even positive integer. Therefore, n can be written as $2k$ for some positive integer $k \geq 1$. Substituting $n = 2k$:

$$\frac{1}{i^{2k}} = \frac{1}{(i^2)^k}$$

Since $i^2 = -1$:

$$= \frac{1}{(-1)^k} = (-1)^{-k} = (-1)^k$$

If k is even (e.g., $n = 4, 8, 12, \ldots$), then $(-1)^k = +1$. If k is odd (e.g., $n = 2, 6, 10, \ldots$), then $(-1)^k = -1$.
Thus, the value is $+1$ or -1.

$$\boxed{\text{A. +1 or -1}}$$

Q5.25.1 MCQ 2025 1M Ans: A ● ○ ○

Let the given matrix be:

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Eigenvalues are found by solving the characteristic polynomial equation $\det(A - \lambda I) = 0$.

$$\begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = 0$$

Evaluating the determinant:

$$(-\lambda)(-\lambda) - (-1)(1) = 0$$
$$\lambda^2 + 1 = 0 \implies \lambda^2 = -1$$
$$\lambda = \pm\sqrt{-1} = \pm i$$

Thus, the eigenvalues are $-\sqrt{-1}$ and $\sqrt{-1}$.

$$\boxed{\text{A. } -\sqrt{-1} \text{ and } \sqrt{-1}}$$

Q5.25.2 MCQ 2025 2M Ans: A ● ○ ○

Let us examine the function in option A:

$$f(z) = e^x(\cos y + i \sin y)$$

Using Euler's formula, $\cos y + i \sin y = e^{iy}$:

$$f(z) = e^x \cdot e^{iy} = e^{x+iy} = e^z$$

Since the complex exponential function e^z is an entire function (analytic everywhere in the complex plane), option A represents an analytic function.

$$\boxed{\text{A.} e^x(\cos y + i \sin y)}$$

Key Points

- Any complex function that can be expressed completely in terms of the single variable $z = x + iy$ without involving its conjugate $\bar{z}$ is analytic where it is differentiable.

Q5.24.1 | MSQ | 2024 | 1M | Ans: A,B,D | ● ● ○

Since $f(x, y) = u + iv$ is an analytic function, its real and imaginary components must satisfy the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{—(i)} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \text{—(ii)}$$

Differentiating equation (i) with respect to x and equation (ii) with respect to y:

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y} \quad \text{and} \quad \frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y \partial x}$$

Adding them together (since $\frac{\partial^2 v}{\partial x \partial y} = \frac{\partial^2 v}{\partial y \partial x}$ for continuous second derivatives):

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \Longrightarrow \text{ Option A is TRUE.}$$

Similarly, differentiating (i) w.r.t y and (ii) w.r.t x shows that v also satisfies Laplace's equation:

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 \quad \Longrightarrow \text{ Option B is TRUE.}$$

Now checking Option D, substitute the CR equations into the expression:

$$\left(\frac{\partial u}{\partial x}\right)\left(\frac{\partial v}{\partial x}\right) + \left(\frac{\partial u}{\partial y}\right)\left(\frac{\partial v}{\partial y}\right) = \left(\frac{\partial v}{\partial y}\right)\left(-\frac{\partial u}{\partial y}\right) + \left(\frac{\partial u}{\partial y}\right)\left(\frac{\partial v}{\partial y}\right) = 0 \quad \Longrightarrow \text{ Option D is TRUE.}$$

$$\boxed{\text{A, B, D}}$$

Q5.21.1 | NAT | 2021 | 1M | Ans: 2.4 to 2.6 | ● ○ ○

The given equation is:

$$(3i + 1)x + (4i + 4)y + 5 = 0$$

Separating the equation into real and imaginary parts:

$$(x + 4y + 5) + i(3x + 4y) = 0 + i0$$

Equating the real and imaginary parts to zero, we get a system of two linear equations:

$$x + 4y + 5 = 0 \quad \text{—(i)}$$

$$3x + 4y = 0 \implies 4y = -3x \quad \text{—(ii)}$$

Substituting equation (ii) into equation (i):

$$x - 3x + 5 = 0$$

$$-2x = -5 \implies x = \frac{5}{2} = 2.5$$

$$\boxed{2.5}$$

Q5.20.1 MCQ 2020 1M Ans: A ● ○ ○

Given complex numbers are $z_1 = 2 + 3i$ and $z_2 = 4 - 5i$.

$$z_1 + z_2 = (2 + 4) + (3i - 5i) = 6 - 2i$$

Hence,

$$(z_1 + z_2)^2 = (6 - 2i)^2 = 36 - 24i + (2i)^2$$
$$= 36 - 24i - 4 = 32 - 24i$$

$$\boxed{\text{A. } 32 - 24i}$$

Q5.18.1 MCQ 2018 2M Ans: A ● ● ○

Given analytic function is $f(z) = x^2 - y^2 + i2xy$. We can express the function in terms of $z = x + iy$:

$$f(z) = (x + iy)^2 = z^2$$

Differentiating $f(z)$ with respect to z:

$$f'(z) = \frac{d}{dz}(z^2) = 2z$$

Substituting $z = x + iy$ back into the derivative:

$$f'(z) = 2(x + iy) = 2x + i2y$$

$$\boxed{\text{A. } 2x + i2y}$$

Q5.17.1 MCQ 2017 1M Ans: D ● ○ ○

Given complex number is $z = x + iy$ and its complex conjugate is $\bar{z} = x - iy$. The product of z and $\bar{z}$ is evaluated as:

$$z\bar{z} = (x + iy)(x - iy) = x^2 - (iy)^2$$
$$z\bar{z} = x^2 - (-1)y^2 = x^2 + y^2$$

$$\boxed{\text{D. } x^2 + y^2}$$

Q5.16.1 MCQ 2016 1M Ans: D ● ○ ○

The given function is:

$$f(z) = \frac{z^2 + 1}{z^2 + 4}$$

Singularities (poles) occur where the denominator is equal to zero:

$$z^2 + 4 = 0 \implies z^2 = -4$$

Taking the square root on both sides:

$$z = \pm\sqrt{-4} = \pm 2i$$

$$\boxed{\text{D. } \pm 2i}$$

Q5.13.1 | MCQ | 2013 | 1M | Ans: C | ● ● ○

Let A be a symmetric matrix, which means $A = A^T$. For any symmetric matrix with real entries, its eigenvalues are always entirely real numbers. Also, eigenvectors corresponding to distinct eigenvalues are orthogonal.

$$\boxed{\text{C. real}}$$

Q5.11.1 | MCQ | 2011 | 2M | Ans: B | ● ● ○

The given integral is:

$$\oint_C \frac{z^2}{z^4 - 1}\, dz$$

Finding poles,

$$z^4 - 1 = (z^2 - 1)(z^2 + 1) = (z - 1)(z + 1)(z - i)(z + i)$$

The singularities are at $z = 1, -1, i, -i$. The given contour is a circle $|z + 1| = 1$, which is centered at -1 with a radius of 1. $z = -1$ lies inside the contour since $|-1 + 1| = 0 < 1$. $z = 1, i, -i$ lie outside the contour.

Hence the integral can be written as,

$$\oint_C \frac{\frac{z^2}{(z-1)(z^2+1)}}{z + 1}\, dz$$

Using Cauchy's Integral Formula where $f(z) = \frac{z^2}{(z-1)(z^2+1)}$ and $a = -1$:

$$\oint_C f(z)\, dz = 2\pi i \cdot f(-1)$$

Evaluating $f(-1)$:

$$f(-1) = \frac{(-1)^2}{(-1 - 1)((1)^2 + 1)} = \frac{1}{(-2)(2)} = -\frac{1}{4}$$

Thus, the value of the integral is:

$$2\pi i \left(-\frac{1}{4}\right) = -\frac{\pi i}{2}$$

$$\boxed{\text{B. } -\pi i/2}$$

Q5.10.1 | MCQ | 2010 | 2M | Ans: A | ● ○ ○

Given equation is $w^3 = 1$, which implies w is a cube root of unity $(1, \omega, \omega^2)$. By elementary properties:

$$w^3 = 1 \quad \text{and} \quad 1 + w + w^2 = 0$$

To find the value of $1 + w + \frac{1}{w}$: Multiplying the numerator and denominator of the last term by w^2:

$$\frac{1}{w} = \frac{w^2}{w^3} = \frac{w^2}{1} = w^2$$

Substituting this back into the expression:

$$1 + w + \frac{1}{w} = 1 + w + w^2 = 0$$

$$\boxed{\text{A. } 0}$$

Q5.08.1 MCQ 2008 1M Ans: B ● ○ ○

The given expression is:

$$\frac{-5+10i}{3+4i}$$

Multiply the numerator and denominator by the complex conjugate of the denominator, which is $3-4i$:

$$= \frac{(-5+10i)(3-4i)}{(3+4i)(3-4i)}$$

$$= \frac{-15+20i+30i-40i^2}{3^2-(4i)^2}$$

$$= \frac{-15+50i+40}{9+16} = \frac{25+50i}{25} = 1+2i$$

$$\boxed{\text{B. 1+2i}}$$

Q5.07.1 MCQ 2007 1M Ans: B ● ● ○

Given complex variable is:

$$z = \frac{\sqrt{3}}{2} + i\frac{1}{2}$$

Expressing z in polar (Euler) form $e^{i\theta}$:

$$z = \cos\frac{\pi}{6} + i\sin\frac{\pi}{6} = e^{i\frac{\pi}{6}}$$

Hence,

$$z^4 = \left(e^{i\frac{\pi}{6}}\right)^4 = e^{i\frac{4\pi}{6}} = e^{i\frac{2\pi}{3}}$$

Converting back to Cartesian form:

$$z^4 = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$$

$$\boxed{\text{B. } -\frac{1}{2} + i\frac{\sqrt{3}}{2}}$$

Q5.05.1 MCQ 2005 2M Ans: A ● ● ○

Given complex function is:

$$w = u + iv = \frac{1}{2}\log(x^2+y^2) + i\tan^{-1}(y/x)$$

This can be rewritten in polar coordinates by substituting $x^2+y^2=r^2$ and $\tan^{-1}(y/x)=\theta$:

$$w = \frac{1}{2}\log(r^2) + i\theta = \log r + i\theta = \log(re^{i\theta}) = \log z$$

The function $w = \log z$ is non-analytic at the origin $z=0$, because the logarithm function is undefined at 0 and its derivative goes to infinity. Thus, it is not analytic at $(x,y)=(0,0)$.

$$\boxed{\text{A. (0,0)}}$$

DEBUG INFO

Complex Variable Analysis

Engineering Mathematics

Total Questions : 151

MCQ : 123

MSQ : 6

NAT : 22